IVF Digital Twin

Version 7.1

An Integrated Multi-Source Ensemble Platform for Stage-Stratified IVF Outcome Prediction

Stochastic Simulation · Neural Network Ensembling · Bayesian Conditional Updating · Unsupervised Phenotype Classification · Diffusion-Based Generative Module · Graph Attention Transformer · Agent Integration (MCP) · Laya

From in vitro to in silico concept

For use with the repository · Please read before running the model

Abstract

Background. Current clinical decision-support tools for in vitro fertilization (IVF) typically address isolated endpoints — oocyte yield, embryo morphology, or implantation probability — and return point estimates without quantifying the substantial biological variability inherent to reproductive medicine. They cannot incorporate stage-by-stage information as the cycle unfolds, do not benchmark a given patient against the spectrum of documented IVF protocol phenotypes, do not integrate independent generative verification of laboratory outcome distributions, and do not contextualise predictions within the observed outcomes of clinically similar patients.

Methods. We constructed a seven-layer probabilistic prediction system (the IVF Digital Twin, v7.1) combining: (i) a seven-stage stochastic Monte Carlo pipeline propagating full probability distributions through controlled ovarian stimulation, oocyte retrieval, maturation, fertilization, blastulation, good-quality blastocyst formation, age-stratified euploidy, and post-thaw survival, whose count distributions were refitted in v7.1 on local clinical data (negative-binomial oocyte yield with a log link, zero-inflated beta-binomial blastulation, beta-binomial blastocyst quality) and which is conditioned exactly on every observed stage count; (ii) a per-transfer pregnancy ensemble combining logistic regression with a KPIScore-based laboratory performance metric, with transfers of one cycle linked by a shared random effect; (iii) a Kolmogorov-Arnold Network and FT-Transformer neural network ensemble (KAT) with isotonic calibration and Bayesian Beta-Binomial posterior synthesis using clinic-specific historical data; (iv) an unsupervised nearest-centroid cluster classifier based on three published IVF protocol phenotypes; (v) a diffusion-based generative module (CSDI Hybrid v3) that independently reconstructs embryological outcome distributions from approximately 15,000 clinical cycles without parametric rate assumptions; (vi) a Graph Attention Transformer (GAT) that constructs a patient-similarity network over 1,172 real clinical protocols and produces a neighbourhood-informed pregnancy probability; and (vii) a Bayesian Evidence Fusion Engine (BEFE) that integrates all upstream layer outputs into a single posterior probability via precision-weighted logit pooling, reported with a model-uncertainty range derived from the remaining scenario spread and a DerSimonian–Laird random-effects pooling of the sources. Version 7.1 runs as a self-contained desktop application with a single computation path, exposed to clinic information systems through an on-premise REST API and to large-language-model (LLM) agents through a Model Context Protocol (MCP) connector; it is coupled to the Laya encoder-classifier as a fast second opinion and serves as the forecasting engine of the laboratory analytics pipeline and the Early Warning System (EWS). All layers were

validated on retrospective cohorts from independent centres, and the frozen model was validated prospectively in silico on the 2026 cohort of Centre E.

Results. On the retrospective validation cohort, the stochastic pipeline reproduced observed median oocyte and blastocyst counts with Spearman correlations of 0.73 and 0.41 respectively. Pregnancy prediction achieved an AUC-ROC of 0.76 with a Brier score of 0.22. The CSDI Hybrid v3 generative module achieved an AUC of 0.661, Brier score of 0.209, and Expected Calibration Error of 0.029 on a hold-out test set of 1,520 cycles. The GAT Graph Transformer achieved an AUC-ROC of 0.632 in standalone cross-validation; the GAT+KAT ensemble achieved approximately 0.658–0.665 with a Brier score of approximately 0.229. The BEFE posterior narrowed the 95% confidence interval from ±18–22 percentage points (raw layer outputs) to ±11–14 percentage points at standard evidence levels, with further narrowing as clinic-specific historical data accumulates. The unsupervised cluster classifier correctly assigned all five reference scenarios to their expected phenotype. In the prospective in silico validation at Centre E (234 fresh transfers with a known outcome, 64 clinical pregnancies; nothing used for prediction was fitted on these data), the DT clinical probability achieved an AUC of 0.702 (95% CI 0.622–0.771), a Brier score of 0.1895 and an observed/expected ratio of 0.94; among transfers with at least one blastocyst the AUC was 0.640 (0.548–0.728). The calibration slope of the raw output was low (0.11) and was corrected by logistic recalibration on the centre's 2023–2025 data (slope 1.21) without any change in AUC.

Conclusions. The IVF Digital Twin v7.1 is, to our knowledge, the first IVF prediction system to integrate stochastic stage-wise simulation, multi-source per-transfer ensemble, calibrated neural-network prediction, Bayesian evidence synthesis, mid-cycle conditional updating, unsupervised phenotype classification, diffusion generative modelling, and patient-similarity graph analysis within a single transparent, auditable framework. All coefficients are traceable to peer-reviewed sources or to documented local fits, the architecture supports prospective external validation, and version 7.1 makes the platform usable by clinic systems and LLM agents without ever exposing patient identity or duplicating the computation.

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1. Introduction

In vitro fertilization (IVF) is among the most technologically sophisticated and clinically uncertain domains of modern medicine. More than 2.5 million treatment cycles are performed worldwide each year, yet live birth rates per cycle remain constrained to 25–35%, and the patient experience is characterized by substantial psychological, physical, and financial burden. The fundamental limitation is not an absence of predictive tools — many exist — but the fragmented, point-estimate-focused, and biologically incomplete character of available systems [1,2].

Existing IVF prediction systems address isolated endpoints. The CDC IVF Success Estimator and the OPIS calculators of McLernon et al. report cumulative pregnancy probabilities over one or more complete cycles but do not decompose the prediction into the underlying biological transitions, nor do they accommodate the substantial inter-cycle variability that characterizes real clinical practice [1–4]. The Orchid Embryo Banking Calculator provides stage-specific point estimates for oocyte yield, blastocyst conversion, and euploidy without confidence intervals or downstream propagation of uncertainty [5]. The Herasight framework introduces a distribution-based stage-structured zero-inflated negative binomial model with conditional updating, achieving well-calibrated prediction intervals, but does not quantify ovarian hyperstimulation syndrome (OHSS) risk, does not model the laboratory stage that embryologists rely upon, and does not benchmark a new patient against the population of IVF protocol phenotypes documented in unsupervised cluster studies [6,7].

The conventional logistic regression and machine-learning models that underpin most commercial calculators produce single scalar probabilities. While statistically convenient, scalar outputs collapse the cascade of conditional events that characterizes an IVF cycle — oocyte retrieval, maturation, fertilization, blastulation, euploidy, transfer, implantation — into one number. They obscure the very questions that patients and clinicians most need answered: what is the full range of plausible outcomes? What is the risk of cancellation? What happens if retrieval underperforms? How will mid-cycle observations change the prognosis?

The Monte Carlo method was selected as the computational foundation of the platform because an IVF cycle is not a deterministic sequence but a chain of stochastic biological filters, and the quantities of genuine clinical interest are therefore not point estimates but full probability distributions. Analytical propagation of uncertainty through such a cascade of coupled, non-linear, integer-valued and conditionally dependent random processes is mathematically intractable in closed form. Monte Carlo simulation resolves this by sampling the entire cascade many thousands of times, allowing the joint distribution of every downstream quantity to emerge directly from the simulated population of synthetic cycles without distributional approximation.

Version 7.0 extended the five-layer architecture of v6.2 by adding two new layers. Layer 6, the Graph Attention Transformer (GAT), constructs a clinical similarity network across the training cohort and produces a neighbourhood-informed pregnancy prediction anchored in the observed outcomes of genuinely similar patients. Layer 7, the Bayesian Evidence Fusion Engine (BEFE), integrates all upstream layer outputs into a single posterior probability with an uncertainty interval, explicit reliability scoring, and out-of-distribution detection. Together, these additions close the two remaining architectural gaps: while Layers 1–5 address the question “what do the clinical parameters suggest?”, Layer 6 addresses “how have patients with profiles like this one actually fared?”, and Layer 7 addresses “how should we weight and combine all these answers, and how confident should we be?”

Version 7.1 adds no new layer. It revises the mechanistic layer and the uncertainty machinery on the basis of local clinical data and opens the platform to integration. The count distributions of Layer 1 were refitted: oocyte yield is a negative binomial with a log link, blastulation a zero-inflated beta-binomial, and good-quality blastocyst formation a beta-binomial whose mean depends on maternal age and on the number of embryos entering the stage. Every blastocyst is transferable without PGT-A, transfers of one cycle share a random effect, observed stage counts are handled by exact conditioning, and the interval beside the headline probability is a model-uncertainty range that narrows as observations arrive. The platform runs as a desktop application with a single computation path that is accessible to clinic information systems through an on-premise REST API and to LLM agents through an MCP connector; it is coupled to the Laya encoder-classifier and serves as the forecasting engine of the laboratory analytics pipeline and the Early Warning System. Finally, the complete model, frozen before the data were scored, was validated prospectively in silico in a fifth centre (Centre E).

2. System Overview

The IVF Digital Twin v7.1 is organized into seven conceptual layers that operate in sequence during a single prediction run (Table 1). Each layer addresses a distinct epistemic question and contributes a distinct piece of information to the final prediction.

Table 1. *Seven-layer architecture of the IVF Digital Twin v7.1*

Layer	Function	Key methods
L1 — Stochastic pipeline	Propagates full probability distributions through seven biological stages of the IVF cycle, from stimulation through post-thaw survival.	Negative-binomial oocyte yield (log link); logistic-binomial MII and 2PN filters; zero-inflated beta-binomial blastulation; beta-binomial quality and age-stratified euploidy; exact conditioning on observed counts. Coefficients from local refits (2026) and from Herasight, Franasiak, Armstrong, Coello.
L2 — Per-transfer ensemble	Combines two independent per-transfer pregnancy estimates: a logistic regression on clinical predictors and a KPIScore-based laboratory performance metric.	FORTUNE-structure logistic; KPIScore (5–25) with per-integer Beta-distributed 95% CIs; logit-weighted ensemble; three-level pregnancy decomposition with a shared transfer random effect ($\sigma = 0.746$).
L3 — Neural network ensemble + Bayes	Calibrated final-layer prediction by a KAN + FT-Transformer ensemble. Bayesian Beta-Binomial posterior fuses NN output, covariate-dependent prior, and clinic-specific retrospective data to produce calibrated CIs.	KAN (3-layer, B-spline activations) + FT-Transformer; isotonic calibration interpolated between step centres; NVSA correction; Beta regression prior; clinic historical batches; per-attempt decay curve.
L4 — Cluster classifier	Independent unsupervised assessment via nearest-centroid assignment to three published IVF protocol phenotypes (Poor, Standard, High responder).	Nearest-centroid in z-score-standardised 18-dimensional space; centroids from 1,556 cycles, k-means k=3; PCA visualization with synthetic clouds.
L5 — Diffusion generative module	Independently reconstructs embryological outcome distributions directly from clinical data. Provides calibrated pregnancy probability and conformal prediction intervals.	CSDI Hybrid v3: Transformer-based diffusion model trained on ~15,000 cycles; LightGBM + Platt classifier; split conformal prediction with coverage guarantees; applicability gate before fusion.
L6 — Graph Attention Transformer	Constructs a patient-similarity graph over the training cohort and produces a neighbourhood-informed pregnancy probability anchored in outcomes of clinically similar patients.	GAT with cosine-similarity edges on 18 clinical features; 3 message-passing layers; multi-task output (pregnancy + PRAI score); trained on 1,172 real protocols; ensemble with KAT: $P = w \cdot P_{GAT} + (1-w) \cdot P_{KAT}$; $w \in [0.25, 0.40]$.
L7 — Bayesian Evidence Fusion Engine (BEFE)	Integrates all upstream layer outputs into a single posterior probability via precision-weighted logit pooling. Produces the final point estimate and a model-uncertainty range. Detects out-of-distribution patients and adjusts reliability accordingly.	Two-level conjugate Gaussian fusion in logit space: Level 1 evidence pooling (KAT, GAT) → P_predictive; Level 2 prior×evidence update (L1/L2 → posterior). Range = scenario spread + DerSimonian–Laird random-effects variance. L5 and L4 modulate reliability. Dual OOD (clinical + embryological subspaces).

All seven layers share a single Monte Carlo loop per patient (N = 2,000 iterations by default in the v7.1 desktop application and API, configurable from 500 to 10,000). Layers L1–L4

operate within each iteration; L5 receives pipeline medians; L6 operates on the final patient feature vector and the static training cohort graph; L7 (BEFE) aggregates the scalar outputs of all upstream layers into a single posterior probability with a model-uncertainty range.

The architecture deliberately combines complementary epistemic sources. The stochastic pipeline encodes population-level biological knowledge from large registries and, since v7.1, from local clinical refits. The FORTUNE-like component reflects an established clinical prediction model derived from over 3,000 patients in randomized trials. The neural network captures complex, non-linear interactions that pre-specified linear models cannot. The Bayesian posterior re-anchors the prediction to the clinic's own historical pregnancy frequency. The cluster classifier provides unsupervised phenotype labelling. The diffusion module provides an entirely data-driven, assumption-free verification from a different modelling paradigm. The graph transformer contextualises the prediction within the observed landscape of genuinely similar patients. Each layer contributes information that the others do not, and the explicit ensembling makes the contribution of each layer auditable.

2.1 Changes in version 7.1

Table 2 summarises what changed between v7.0 and v7.1. The neural-network weights (KAN, FT-Transformer, GAT, CSDI) were not retrained; every change concerns the mechanistic layer, the transfer scenario, the uncertainty range, calibration and deployment. The coefficients actually used by v7.1 are the module-level constants of src/embryology.py (refit of 11 September 2026); earlier coefficient files are kept as provenance only.

Table 2. *Summary of changes from v7.0 to v7.1*

Component	v7.0	v7.1
Stage 1 — oocytes	Zero-inflated NB: logistic cancellation component + ART-ONE predictor with attempt term; truncation at 45	NB with log link and no structural zero (not identifiable); refitted on 687 stimulated cycles; truncation at 50 (Section 4.0)
Stage 4 — blastocysts	Binomial; rate 0.70 − 0.012·max(0, age − 40) with Normal (SD 0.06) perturbation	Zero-inflated beta-binomial; age hinge at 35 and ln(min(2PN, 5)) term; $\kappa = 7.82$, $\pi_0 = 0.0095$; n = 7,919 (Section 4.3)
Stage 5 — good blastocysts	Binomial with Beta mixing (concentration 10); 0.78 − 0.008·max(0, age − 35)	Beta-binomial; age hinge at 35 and ln(Bl) term; $\kappa = 4.13$ (Section 4.4)
Stage 6 — euploidy	Binomial on good blastocysts with smooth age lookup	Beta-binomial ($\kappa = 6$) on good blastocysts with age-band rates (Section 4.5)
Transfer scenario	Warmed euploid embryos	Without PGT-A every blastocyst is transferable: good quality first, then fair quality with OR 0.51; entered euploid count → PGT-A scenario (Section 4.6)
Several transfers per cycle	Independent	Shared logit-normal random effect, $\sigma = 0.746$ (Section 5.4)
Mid-cycle observations	Observed stage replaced by a point mass	Exact forward filtering / backward sampling of the whole stage chain (Section 8)
KAT calibration	Venn-Abers wrapper	Isotonic function interpolated between step centres; batch-independent PLE encoding (Sections 6.4, 6.9)

Component	v7.0	v7.1
BEFE interval	95% CI from pooled precision	Model-uncertainty range: scenario spread + DerSimonian–Laird random-effects variance; coverage not validated (Section 13.2.3)
Clinic adaptation	Beta-Binomial historical batches	Unchanged, plus centre intercept calibration with inclusion criteria and model fingerprint (Section 7.7)
Outcome definition	Clinical pregnancy confirmed by ultrasound	Clinical pregnancy confirmed by ultrasound, including confirmed ectopic pregnancy
Deployment	Research script and Streamlit application	Desktop application with a single computation path; REST API v1; MCP connector; Laya; analytics pipeline and EWS (Part VI)
Validation	Retrospective cohorts	Plus multicentre in silico cohort and prospective in silico validation in Centre E (Sections 14.7–14.8)

3. Monte Carlo Engine and Risk Quantification

The stochastic pipeline models an IVF cycle as a sequence of seven probabilistic stages. Each stage takes the output distribution of the preceding stage as its input, applies a calibrated stochastic filter, and passes a new distribution to the next stage. The complete cascade runs inside the Monte Carlo engine for N iterations per patient.

3.1 Sample size and convergence

Convergence testing confirmed that mean and 95th-percentile estimates of oocyte count, blastocyst count, and per-transfer pregnancy probability stabilize to within 0.5 percentage points by N = 2,000. In v7.1, N = 2,000 is the default of the desktop application, the batch calculator and the REST API (configurable from 500 to 10,000; a reproducible seed is stored with every result), while the research core retains N = 5,000 for offline runs. Computation time is negligible on a standard workstation. Because observed stage counts are handled by exact conditioning (Section 8) rather than by discarding inconsistent iterations, every iteration remains usable when observations are entered.

3.2 Explicit risk metrics

3.2.1 Ovarian hyperstimulation syndrome (OHSS)

OHSS risk is estimated per iteration from the number of retrieved oocytes using published ESHRE threshold criteria [21,22]:

- Mild OHSS risk: P(COC > 15) estimated empirically across iterations.
- Moderate-to-severe OHSS risk: P(COC > 20), with additional elevation flag for AMH > 3.5 ng/mL (high-response biochemical marker).
- Recommended trigger: GnRH-agonist trigger and freeze-all strategy when P(COC > 20) > 40% or AMH > 3.5 ng/mL, per current ESHRE guidance.

Under the v7.1 oocyte model (Section 4.0), P(COC > 15) and P(COC > 20) are 22.6% and 7.9% for a standard responder (age 35, AMH 2.5 ng/mL, AFC 15) and 70.7% and 49.9% for a high responder (age 28, AMH 4.5 ng/mL, AFC 28).

3.2.2 Empty cycle risk

The probability of an empty cycle — defined in v7.1 as no transferable embryo: no warmed blastocyst without PGT-A, or no warmed euploid blastocyst once a euploid count has been entered (PGT-A scenario, Section 4.6) — is computed as the empirical fraction of Monte Carlo iterations without a transferable embryo. This is reported explicitly as P(empty) and constitutes a clinically critical safety endpoint that single-scalar models are structurally incapable of reporting. An observed zero at the oocyte, MII, 2PN or blastocyst stage closes the cycle; zero good-quality blastocysts alone does not.

3.3 Visualization of distributional outputs

The engine generates full distributional outputs for every stage: violin plots, 95% credible interval ribbons, and probability mass functions. The PCA scatter plot for cluster classification (Section 10) and the patient-similarity network for the GAT layer (Section 12) are also generated within the same Monte Carlo run, ensuring that all visualizations are mutually consistent and derived from the same set of synthetic cycles.

4. Stochastic Pipeline (Layer 1)

The stochastic pipeline models an IVF cycle as a sequence of seven probabilistic stages. Each stage takes the output distribution of the preceding stage as its input, applies a calibrated stochastic filter, and passes a new distribution to the next stage. The complete cascade runs inside the Monte Carlo engine (Section 3) for N iterations per patient.

4.0 Stage 1 — Retrieved oocytes

The number of retrieved oocytes (OKK) is modelled in v7.1 with a negative-binomial (NB) distribution with a log link. Oocyte yields are right-skewed with heavier tails than the Normal, particularly among high responders: variance grows quadratically with the mean rather than linearly. This overdispersion is formally accommodated by the NB.

The zero-inflation component of v7.0 (a logistic cancellation probability, logit(P_zero) = −3.2 + 0.40·z_age − 0.60·z_AMH − 0.80·z_AFC) has been removed. When the model was refitted on 687 stimulated cycles with at least one retrieved oocyte (one centre, July 2025 – 2026; five-fold cross-validation grouped by patient), blank oocyte counts were explained by the NB zero together with a constant 5.8% rate of unrecorded retrievals, so that a separate structural-zero component was not identifiable. The model was fitted as a zero-truncated NB on the positive counts; at prediction a zero yield remains possible and its probability is the NB zero itself.

The mean number of retrieved oocytes is:

$$\mu_1 = \exp(c_0 + c_{AMH} \cdot \ln(AMH + 0.1) + c_{AFC} \cdot \ln(AFC + 1) + c_{age} \cdot (age - 35))$$

$$OKK \sim NB(\theta, \theta / (\theta + \mu_1)), \quad Var(OKK) = \mu_1 + \mu_1^2 / \theta, \quad \text{clipped to } [0, 50]$$

with $c_0 = 0.0598$, $c_{AMH} = 0.2684$, $c_{AFC} = 0.7689$, $c_{age} = -0.00393$ and dispersion $\theta = 5.77$. AMH is expressed in ng/mL, AFC is the antral follicle count, and age is the patient's age in years. The log-AMH and log-AFC structure of the ART-ONE formula is retained, but the attempt-number term was dropped, and BMI has no term because it was not recorded in the fitting data. The refit corrects an under-estimation of yield by the former linear predictor in low-reserve patients (for AFC < 5: 1.4 predicted vs 2.7 observed oocytes). The stage includes an upper truncation at 50 oocytes to suppress biologically implausible extremes; the exact conditional calculation (Section 8) keeps the NB tail mass above the cap and extends the support to 100 when a higher count is actually observed.

Representative outputs: standard responder (age 35, AMH 2.5, AFC 15): mean 11.6, SD 5.9, P(OKK = 0) 0.2%; poor responder (age 42, AMH 0.5, AFC 6): mean 4.0, SD 2.6, P(OKK = 0) 4.7%; high responder (age 28, AMH 4.5, AFC 28): mean 21.9, SD 10.2.

4.1 Stage 2 — Mature (MII) oocytes

Maturation is modelled as a binomial filter applied to the simulated oocyte count, with a per-oocyte success probability obtained from a logistic regression. The coefficients are taken verbatim from the Herasight simplified maturity model (Table A1, column 3), fit on 90,479 HFEA cycles. For our clinical context (stimulation = 1, ICSI, unexplained infertility) the linear predictor reduces to:

$$\text{logit}(p_MII) = 2.4665 + 0.005 \cdot age - 0.782 \cdot stim + 0.24 \cdot AMH - 0.069$$

This collapses to a population-mean maturity rate of approximately 90.7 percent at the HFEA average age and AMH, which is precisely the empirical mean reported by the source registry. The MII count for iteration i is then drawn from Binomial (OKK_i, p_MII). The choice of fixed-coefficient logistic over earlier beta-binomial heuristics eliminates spurious

overdispersion and ensures that the maturity stage is traceable to a single peer-reviewed source. This stage is unchanged in v7.1.

4.2 Stage 3 — Two-pronuclear zygotes (2PN)

Fertilization is modelled analogously. The per-MII success probability is taken from the Herasight simplified fertilization model (Table A1, column 5), with the same domain reduction (ICSI, stimulation, unexplained infertility):

$$\text{logit}(p_fert) = 1.1678 + 0.004 \cdot age - 0.303 \cdot stim - 0.051$$

This yields a population-mean fertilization rate of 72.3 percent at the HFEA mean age, matching the empirical value. The conventional-IVF coefficient (−0.296) is omitted because all cycles in our clinic use ICSI. The 2PN count is drawn from Binomial (MII_i, p_fert). Stages 2 and 3 together replace earlier expert-driven heuristic models with peer-reviewed registry-derived coefficients while preserving the upstream stochasticity of the oocyte yield distribution. This stage is unchanged in v7.1.

4.3 Stage 4 — Blastocysts

Blastocyst formation is modelled in v7.1 as a zero-inflated beta-binomial (ZIBB) filter. Earlier versions used a binomial filter whose rate was held stable through age 40 and declined thereafter (p_blast = clip(0.70 − 0.012·max(0, age − 40), 0.30, 0.75), calibrated against Romanski et al. and Sainte-Rose et al.), with a Normal per-iteration perturbation of SD 0.06. The v7.1 refit on 7,919 culture-observed cycles of the 15,000-cycle protocol database (day-5 to day-7 transfers recorded with zero blastocysts were excluded because their blastocyst fields were partly unfilled) showed that blastulation depends on maternal age, with a hinge at 35 years, and on the number of zygotes in culture:

$$\text{logit}(\mu_4) = 0.6143 - 0.01696 \cdot \max(0, age - 35) + 0.01959 \cdot \ln(\min(2PN, 5))$$

$$p_{4,i} \sim \text{Beta}(\mu_4\kappa_4, (1 - \mu_4)\kappa_4), \quad \kappa_4 = 7.82$$

$$Bl \mid 2PN \sim \pi_0 \cdot \delta_0 + (1 - \pi_0) \cdot \text{Binomial}(2PN, p_{4,i}), \quad \pi_0 = 0.0095$$

The beta mixing replaces the Normal perturbation of v7.0: it represents inter-cycle and inter-laboratory variability that fixed coefficients cannot capture, with support confined to [0, 1]. The structural zero π_0 represents complete developmental arrest of a cohort (about 1% of cycles), which a beta-binomial alone under-represents. The central blastulation rate is 65.6% up to age 35 (five or more zygotes), 63.7% at age 40 and 61.7% at age 45. The absence of blastocysts is an explicit outcome of the stage.

4.4 Stage 5 — Good-quality blastocysts

The proportion of blastocysts graded as good quality (Gardner 3BB or higher) is modelled in v7.1 as a beta-binomial filter whose mean depends on maternal age, with a hinge at 35 years, and on the number of blastocysts formed:

$$\text{logit}(\mu_5) = 0.0882 - 0.0635 \cdot \max(0, age - 35) + 0.3572 \cdot \ln(Bl)$$

$$Good \mid Bl \sim \text{BetaBinomial}(Bl, \mu_5\kappa_5, (1 - \mu_5)\kappa_5), \quad \kappa_5 = 4.13$$

As in Stage 4, the per-iteration rate is drawn from a Beta distribution around the central mean, capturing grading variability across embryologists and laboratories; the concentration fitted on local data (4.13) is lower than the value of 10 used in v7.0, reflecting a wider between-cycle spread of morphological grading. The positive count term reflects the observation that cycles producing more blastocysts also produce a larger share of good-quality ones. Up to age 35 the central good-quality fraction is 52.2% for a single blastocyst and 64.2% for four blastocysts; at age 45 it is 36.7% and 48.7%. In v7.1 blastocyst quality

determines the order of transfer and the per-transfer odds (Section 4.6), not whether a blastocyst is transferable.

4.5 Stage 6 — Euploid embryos

Aneuploidy is the single largest determinant of implantation failure and miscarriage, and its prevalence rises sharply with maternal age. The age-specific euploidy rate is taken from the combined evidence of Franasiak et al. (15,169 trophectoderm biopsies) and Armstrong et al. (86,208 cycles) and from our routine practice with 4,863 treatment cycles, synthesized into an age-stratified central value.

To convert this single central value into a per-iteration random rate, the central value is used as the mean of a Beta (α, β) distribution with concentration κ = 6 (α = mean·κ, β = (1 − mean)·κ). This formulation introduces appropriate variability at the embryo level while preserving the age-stratified central tendency. For each Monte Carlo iteration the simulated euploid count is bounded above by the good-blastocyst count:

$$\text{Euploid} \mid \text{Good} \sim \text{BetaBinomial}(\text{Good}, e(\text{age})\cdot\kappa, (1 - e(\text{age}))\cdot\kappa), \quad \kappa = 6$$

where e(age) is the age-band euploid fraction used by v7.1: 0.70 below 30 years, 0.65 at 30–34, 0.55 at 35–37, 0.35 at 38–39, 0.18 at 40–41 and 0.10 from 42 years. This is currently the most data-rich calibration source available for this parameter. The euploid stage restricts transfer only in the PGT-A scenario (Section 4.6); without PGT-A it is reported but does not limit which blastocysts can be transferred.

4.6 Stage 6b — Post-thaw survival and the transfer scenario

Post-thaw blastocyst survival is modelled as a binomial filter with the fixed survival probability of 95% drawn from Coello et al. (2021) [20], reflecting vitrification survival rates in contemporary high-quality IVF laboratories. This stage translates the blastocyst count into the number of embryos available for transfer in a frozen embryo replacement cycle.

Transfer scenario (v7.1). Without PGT-A every blastocyst is transferable. Good-quality blastocysts are transferred first, followed by fair-quality ones, each after warming:

$$n_{good} \sim \text{Binomial}(\text{Good}, 0.95), \quad n_{fair} \sim \text{Binomial}(\text{Bl} - \text{Good}, 0.95)$$
$$\text{logit}(p_{fair}) = \text{logit}(p) - 0.666 \quad \text{(adjusted OR 0.51, 95\% CI 0.35–0.75)}$$

The fair-quality odds ratio was estimated on 2,975 day-5 single blastocyst transfers (399 with fair-quality embryos only), adjusted for age and ln(2PN). Zero good-quality blastocysts alone therefore does not close the cycle. When a euploid count is entered, the model switches to the PGT-A scenario, in which only warmed euploid embryos are transferable: $n_{good} \sim \text{Binomial}(\text{Euploid}, 0.95)$ and $n_{fair} = 0$. The headline probability is estimated over the scenarios in which a transfer is possible; the per-cycle probability also accounts for the absence of a transfer and for all available embryos.

4.7 Summary of the stochastic pipeline

The seven-stage cascade transforms the patient's pre-cycle inputs (age, AMH, AFC, BMI) into a complete distribution over every embryological endpoint: oocyte yield, MII count, 2PN count, blastocyst count, good-quality blastocyst count, euploid count, and warmed transferable embryos. Every quantity is a random variable, not a point estimate, and every downstream quantity is jointly consistent with every upstream quantity within each Monte Carlo iteration. Quantities that the simulation does not observe but which the downstream networks require (punctured follicles, day-5 embryos, cryopreserved embryos) are derived

with the definitions used in their training data — for example, follicles = round(OKK / 0.846), the median ratio in the 15,000-protocol database. The result is a principled, uncertainty-aware model of the full biological cascade whose v7.1 parameters are traceable either to published registries or to documented local refits.

5. Pregnancy Probability — Multi-Source Ensemble (Layer 2)

Layer 2 combines two independent per-transfer pregnancy estimators using a logit-scale weighted average. Both estimators are computed per Monte Carlo iteration, ensuring that the ensemble inherits the full distributional output of Layer 1.

5.1 Per-transfer probability

The first per-transfer estimator follows the structure of the FORTUNE prediction model recently published by IVIRMA-NJ, applied to euploid warmed blastocyst transfers in top-decile SART clinics. The model expresses the log-odds of clinical pregnancy as a linear function of standardized patient covariates:

$$\text{logit}(p_F) = 0.40 - 0.55 \cdot z_{age} + 0.15 \cdot z_{AMH} - 0.20 \cdot z_{BMI}$$

where $z_{age} = (age - 36.3)/5.5$, $z_{AMH} = (\log(AMH + 0.1) - \log(2.1))/1.2$, and $z_{BMI} = (BMI - 24.0)/4.2$. The intercept of 0.40 corresponds to a baseline per-transfer pregnancy probability of approximately 60 percent at the population mean, anchored to SART top-decile euploid-transfer live birth rates [25]. The age slope is intentionally weaker than in age-only models because much of the age effect is already encoded upstream through stage-by-stage euploidy decline. A small Gaussian random effect (SD = 0.07 on the logit scale, equivalent to ±3 percentage points around the central rate) is added per Monte Carlo iteration to reflect biological variability that fixed coefficients cannot capture.

5.2 KPIScore-based per-transfer probability

The second per-transfer estimator is derived from KPIScore, a composite five-component laboratory KPI score. The score is calculated per Monte Carlo iteration using the simulated laboratory outputs of that specific iteration, so it is fully integrated with upstream stochasticity rather than averaged across runs. Each of five components contributes 1, 3, or 5 points:

Component	Score 1	Score 3	Score 5
Female age (years)	≥ 40	37–39	≤ 36
AMH (ng/mL)	< 1	1.0–1.99	≥ 2
MII oocytes (simulated)	≤ 3	4–6	≥ 7
Fertilisation rate (%)	< 50	50–65	> 65
Good blastocysts (simulated)	0	1–2	≥ 3

Total scores range from 5 (worst) to 25 (best). Each integer score is mapped to a clinical implantation probability with a published 95% confidence interval. At each Monte Carlo iteration, the per-iteration KPI probability is sampled from a Beta distribution whose parameters are moment-matched to the score's 95% CI, ensuring that the KPI probability inherits both the patient's central score and the score's published uncertainty interval.

5.3 Logit-scale ensemble combination

The two per-transfer probability vectors (p_F and p_{KPI}) are combined per Monte Carlo iteration on the logit scale:

$$\text{logit}(p_{ens}) = (1 - w) \cdot \text{logit}(p_F) + w \cdot \text{logit}(p_{KPI})$$

The KPI weight w defaults to 0.5 (equal weighting). Logit-scale averaging is preferred over probability-scale averaging because it respects the natural odds-ratio metric of probabilities and avoids systematic bias near the 0/1 boundaries.

5.4 Three-level pregnancy decomposition

The platform reports pregnancy at three explicitly ordered levels to resolve the apparent paradox that for poor responders, per-transfer rate can exceed cumulative cycle probability. Strict mathematical ordering is guaranteed at every level: per-transfer is always less than or equal to cumulative-if-viable (because additional transfers can only increase the probability of at least one success), and overall is always less than or equal to cumulative-if-viable (because P(viable) ≤ 1). A defensive max() safeguard at the algorithmic level prevents rare Monte Carlo noise from violating the ordering. The decomposition has three immediate clinical applications. First, it lets the clinician answer the per-transfer question directly when counselling a patient who is about to receive an embryo. Second, it lets the clinician quote a meaningful cycle-success number that does not depend on whether the patient happens to have viable embryos. Third, it makes the role of the empty-cycle risk transparent in the overall figure:

Level	Conditional event	Mathematical definition
[1] Per-transfer rate	Given that a transfer takes place	p_ensemble (central scalar)
[2] Cumulative-if-viable	Conditional on cycle producing ≥ 1 transferable embryo	$E_z[1 - \Pi_k(1 - p_k(z)) \mid n_tx \geq 1]$
[3] Overall cycle success	From start of stimulation, including empty-cycle risk	P(viable) × Cumulative-if-viable

Strict mathematical ordering is guaranteed: [1] ≤ [2] and [3] ≤ [2] at every iteration.

Correlated transfers (v7.1). Transfers of embryos from one retrieval are not independent events. In v7.1 the cumulative-if-viable level is computed with a shared logit-normal random effect across the transfers of one cycle:

$$\text{logit}\, p_k(z) = \ell(p_k) + \sigma z, \quad z \sim N(0, 1), \quad \sigma = 0.746$$

$$P(\geq 1 \text{ pregnancy} \mid n_{good}, n_{fair}) = E_z[1 - (1 - p_{good}(z))^{n_good} \cdot (1 - p_{fair}(z))^{n_fair}]$$

where $\ell(p)$ is the logit location whose mean over z equals the marginal per-transfer probability p, so that every single transfer keeps its marginal probability and a cycle with one available embryo returns exactly p; the expectation is evaluated by 32-point Gauss–Hermite quadrature. σ was estimated on 1,703 retrievals in which the outcome of every transfer was recorded (2,406 transfers, 461 retrievals with two or more): 95% profile CI 0.29–1.13, latent intra-cycle correlation 0.145; adjustment for age and blastocyst count leaves 0.732, and the single-embryo transfer subset gives 0.681. A structural never-implanting share was not identifiable and is not modelled. Each additional embryo therefore adds less than under independence: for p = 0.40 the probability of at least one pregnancy is 61.4% after two transfers and 73.9% after three, compared with 64.0% and 78.4% for independent transfers. The field “Planned transfers” limits the calculation to the stated number of sequential transfers and labels the per-cycle probability accordingly; it does not affect the per-transfer probability or the neural-network inputs (the KAT feature “transferred embryos” remains one, as in training).

5.5 Discrete pregnancy count distribution

Beyond the binary outcome, the platform reports the full discrete distribution over the number of clinical pregnancies achieved during sequential transfer of all available

transferable embryos. In v7.1 each Monte Carlo iteration first draws its shared effect z_i, and the transfer outcomes of that iteration are drawn with the probabilities $p_k(z_i)$, which keeps the sampled counts consistent with the cumulative probabilities of Section 5.4. The probability of achieving at least k pregnancies is computed empirically as the fraction of iterations meeting that threshold.

6. Neural Network Final Layer — The KAT Ensemble (Layer 3)

The third architectural layer is a calibrated neural-network ensemble called KAT, for Kolmogorov-Arnold network + FT-Transformer. Positioned at the end of the prediction pipeline, it consumes all 18 simulated and observed laboratory variables from every Monte Carlo iteration and produces a per-iteration pregnancy probability that is subsequently NVSA-adjusted, Bayesian-synthesized, and projected onto the per-attempt decay curve.

6.1 Rationale for a neural-network final layer

The pipeline up to and including Layer 2 is composed of linear, log-linear, or additive components. While transparent and well-validated, this structure cannot capture complex interactions among predictors — for example, the prognostic effect of three good-quality blastocysts may depend non-linearly on the patient's age, attempt number, and fertilization rate together. A neural-network layer provides a principled way to capture such interactions without pre-specification.

Two architectures with complementary strengths are ensembled. Kolmogorov-Arnold Networks (KANs) are based on the theorem that any multivariate continuous function can be represented as a composition of univariate functions, giving them an inductive bias toward smooth, interpretable decompositions that generalize well in small-data regimes. Feature-Tokenizer Transformers (FT-Transformers) tokenize each feature into a learned embedding and apply self-attention across the resulting token sequence, giving them a strong bias toward capturing cross-feature interactions explicitly.

6.2 KAN architecture

The KAN component is implemented as a three-layer perceptron with the canonical KAN structure [1, 10, 1]: a single-output logistic head on top of ten learnable activation functions parameterized by B-splines. The B-spline grid size is 5 and the spline order k is 3. Optimization uses the Adam algorithm with both L1 and entropy regularization terms on the activation functions to encourage parsimony. The loss function is binary cross-entropy with logits (BCEWithLogitsLoss).

6.3 FT-Transformer architecture

The FT-Transformer component is implemented through the FT-Transformer classifier of the rtdl-revisiting-models package, following the original specification of Gorishniy et al. [29]. Each of the 18 input features is first projected into a learnable embedding (the feature tokenizer), and the resulting token sequence is processed by a stack of transformer blocks with multi-head self-attention. A classification head produces the pregnancy probability.

6.4 Ensemble combination and calibration

The KAN and FT-Transformer are combined by a learned weighted average of their probability outputs, with two non-negative scalar weights (initialized at 0.5 each, learned jointly with the base models). In v7.0 the combined output was passed through a Venn-Abers conformal calibration wrapper (venn-abers package v1.4.5), which transforms the

raw output into a probability interval with distribution-free coverage guarantees [30,31]. In v7.1, by decision of the platform owner, the combined output is calibrated by an isotonic function fitted on the development data and interpolated linearly between the centres of its steps. Interpolation removes the plateaus of a step function while preserving monotonicity, so that two patients with slightly different raw scores no longer receive identical probabilities. The change of interpolation is not a claim of improved clinical accuracy, and the exact provenance of the shipped isotonic artefact is not fully reconstructed. The calibrated value enters BEFE from the same conditional scenario representation that is used in the report.

Following the operational mapping established in our development workflow: the number of inseminated oocytes is taken to equal MII (every MII oocyte is inseminated); the number of cleaving embryos is taken to equal 2PN (every fertilised zygote enters culture); the number of day-5 embryos is also taken to equal 2PN (all cultured embryos are evaluated on day 5 — no D3 transfer or cryopreservation); the number of cryopreserved embryos equals the good-quality blastocyst count (since only good-quality blastocysts are vitrified in our protocol); and the number of transferred embryos is fixed to one, reflecting the elective single embryo transfer (eSET) policy used throughout the validation cohort.

6.5 Input features

The KAT ensemble takes 18 features in a fixed order, exactly matching the order used during training:

Group	Features
Patient demographics	Female age (years); IVF attempt number
Ovarian response	Follicle count on retrieval day; retrieved oocyte (COC) count
Embryological counts	MII oocytes (inseminated); 2PN zygotes; cleaving embryos on day 3; total blastocysts; good-quality blastocysts; day-5 embryos; cryopreserved embryos; transferred embryos
Laboratory KPIs	Fertilisation rate (2PN ÷ MII); cleavage rate; blastocyst formation rate; TGBDR (good blasts ÷ 2PN); oocyte retrieval efficiency (COCs ÷ follicles)
Composite score	KPIScore (5–25), computed inside the NN using the simulated iteration values

In v7.1 the KPIScore passed to KAT, GAT and CSDI is computed exactly as recorded in their training data: the ovarian-reserve component uses the follicle count (> 15: 5 points; 8–15: 3; < 8: 1) in place of AMH, and the MII component uses the bands ≤ 3, 4–7 and > 7. The KPIScore of Layer 2 (Section 5.2) keeps its AMH-based definition.

6.6 NVSA correction

Raw NN output can drift from the empirically calibrated KPIScore-based probabilities for unusual patient profiles outside the dense region of the training distribution. The NVSA (Neural-Validation Scoring Adjustment) correction anchors the NN output to the KPIScore prior by computing a correction factor per Monte Carlo iteration, clipped to the interval [1/1.5, 1.5]. This preserves the NN's relative ranking of patients while pulling extreme probabilities back toward the KPIScore-anchored prior.

6.7 Per-attempt probability decay curve

The platform runs the neural network forward pass six times per Monte Carlo iteration, varying the attempt number from 1 to 6 while keeping all other features fixed at their per-

iteration simulated values. The result is a per-attempt probability decay curve that quantifies the diminishing returns of sequential cycles. This curve is directly actionable for counselling patients about the recommended number of treatment cycles.

6.8 Graceful fallback

If the KAN weights, FT-Transformer artefact, or calibration files are not present, or if the required Python dependencies are unavailable, the NN layer is automatically bypassed and the FORTUNE+KPI ensemble (Section 5) is used as the final prediction. This design lets users run the platform without the full deep-learning stack while retaining all other features.

6.9 Feature-encoding consistency fix (v7.1)

A review of KAT in September 2026 found that the piecewise-linear encoding (PLE) of the FT-Transformer derived its thresholds from the maximum of each feature in the current batch, so that one patient scored alone and the same patient scored within a batch could receive different probabilities. The bundled mambular library was patched to keep the thresholds of the trained tree. On nine synthetic profiles the discrepancy between single and batch scoring was up to 0.12047 before the fix and below 1.1×10^{-7} after it (tolerance 10^{-6}). This is a check of software consistency, not of pregnancy accuracy. On 1,606 rows of the reconstructed validation part of the 15,000-protocol database the outputs before and after the fix coincided: Brier score 0.1594 and ECE (10 equal-width bins) 0.0279 — a re-evaluation of an existing cohort, not an independent external validation. The KAN, FT-Transformer weights and the isotonic artefact were not changed. Because the patched file is part of the model fingerprint (Section 16.2), centre calibrations made before the fix must be recomputed.

7. Bayesian Posterior Synthesis and Clinic-Specific Calibration

The NVSA-adjusted neural network probability is a point estimate. To convert it into a calibrated posterior distribution that explicitly incorporates the clinic's own historical pregnancy frequency, the platform applies a Beta-Binomial conjugate update. This section documents the update mechanism, its three input streams, the effect of accumulating clinic-specific data on confidence interval width, worked numerical examples, and the centre intercept calibration added in v7.1.

7.1 Beta-Binomial conjugate framework

The posterior for the per-transfer pregnancy success rate p is computed via Beta-Binomial conjugate updating. If the prior belief about p is Beta(α, β) and we observe s successes in t Binomial trials, the posterior is Beta(α + s, β + t − s). The posterior mean is (α + s)/(α + β + t) and the 95% credible interval is obtained from the inverse Beta CDF at the 0.025 and 0.975 quantiles. This formulation is computationally exact, closed-form, and supports sequential updating as new clinical batches become available.

7.2 Covariate-dependent prior

Versions 7.0 and 7.1 use a covariate-dependent Beta regression prior. The prior mean is set to the FORTUNE-based per-transfer probability for the specific patient — computed from her age, AMH, and BMI before any laboratory-phase information enters the model. The prior precision is fixed at κ = 20 pseudo-observations, diffuse enough to be overridden by real clinical data while preventing degenerate posteriors when clinic data is sparse.

The practical consequence: for a standard responder (age 35, AMH 2.5) the prior mean is 64.7%; for a poor responder (age 42, AMH 0.5) it is 40.8%; for a high responder (age 28, AMH 4.5) it is 80.6%. After updating with the same clinical batches and neural network evidence, these three patients converge toward the clinic's historical rate from different starting positions, yielding posterior means of 44.2%, 34.5%, and 46.8% respectively.

7.3 Three sources of evidence

The Bayesian update synthesizes three distinct, epistemically non-redundant inputs:

1. The covariate-dependent Beta prior, based exclusively on pre-laboratory demographic and reserve predictors (age, AMH, BMI), epistemically independent of the laboratory-phase evidence that arrives later.
2. The clinic's own historical outcomes, entered as observed Binomial data (successes and trials). By default, nine retrospective batches totalling 154 successes in 382 transfers are used; the user can supply alternative batches at runtime. As the clinic accumulates cycles, this component increasingly dominates the posterior. In automatic research scoring through the API (Section 18.3) the clinic prior is pinned per batch and is part of the model fingerprint.
3. The NVSA-adjusted NN probability for the current patient, incorporated as 100 pseudo-observations. The weight of NN evidence relative to the prior and real data is controlled by this hyperparameter and can be adjusted by the user.

7.4 Effect of clinic-specific data on confidence interval width

The most practically important property of the Bayesian posterior is the progressive narrowing of the 95% credible interval as the clinic accumulates historical outcome data. This narrowing directly reflects the platform's increasing knowledge of the treating centre's own performance envelope.

The mechanism is explicit in the Beta-Binomial framework: the posterior variance $\mathrm{Var}(p) = (\alpha + s)(\beta + t - s) / [(\alpha + \beta + t)^2 \cdot (\alpha + \beta + t + 1)]$ decreases monotonically as t increases. The width of the 95% credible interval therefore narrows continuously as more clinic data are provided.

Table 3. *Effect of clinic-specific data volume on 95% CI width (central scenario: posterior mean ≈ 45%)*

Clinic data (transfers)	95% CI width	CI example	Clinical interpretation
No clinic data (prior only)	±22–26 pp	[23%, 69%]	Wide; dominated by population prior. Use cautiously for individual counselling.
50 transfers	±18–22 pp	[26%, 64%]	Moderate narrowing. Clinic rate beginning to anchor the posterior.
150–382 transfers (default)	±11–14 pp	[29%, 61%]	Clinically actionable. Posterior reliably reflects local performance.
300 transfers	±10–12 pp	[33%, 57%]	Narrow. Clinic's historical rate dominates; NN prior contributes modestly.
600+ transfers	±7–9 pp	[36%, 54%]	Tight. The posterior is largely determined by the clinic's own experience.

Two observations follow from Table 3. First, the default nine-batch configuration (154 successes in 382 transfers) already places the platform in the “clinically actionable” regime where the 95% CI is narrow enough to support individualized counselling. Second, as the

platform is used prospectively and clinic data is updated, the CI narrows continuously — this is the dynamic calibration property that distinguishes a Bayesian framework from any fixed-parameter model.

7.5 Worked numerical example

A 38-year-old patient, AMH 1.2 ng/mL, AFC 8, BMI 22; clinic historical rate 42% (154 successes / 382 transfers).

- Prior: FORTUNE prior mean = 46.1%; precision $\kappa = 20$. Beta(9.2, 10.8).
- NN evidence: NVSA-adjusted KAT output = 39%. Incorporated as 100 pseudo-observations: Beta contribution (+39, +61).
- Clinic data: +154 successes, +228 failures.
- Posterior: Beta(202.2, 299.8). Mean = 40.3%. 95% CI = [35.6%, 45.1%]. Width = 9.5 pp.

Without clinic data (prior + NN only): mean = 41.8%, 95% CI = [28.4%, 55.8%], width = 27.4 pp. Incorporating 382 clinic transfers narrows the CI nearly three-fold.

7.6 Practical guidance for clinic data entry

The platform accepts clinic historical data in two forms: (a) retrospective batches entered as (successes, trials) pairs at startup; and (b) prospective per-patient updates entered as each cycle is completed. For new clinics, the nine default literature-derived batches serve as a universal prior; these should be replaced with clinic-specific data as soon as a local dataset of 50–100 completed transfer cycles is available. The number of NN pseudo-observations (default: 100) can be reduced if the clinic distrusts the neural network output for its specific population.

7.7 Centre intercept calibration (v7.1)

Version 7.1 adds a second, explicit adaptation step: a shift of the intercept of the logit of the per-transfer probability, fitted on the centre's own outcomes and applied before transfer. Eligible cycles use own oocytes, a single blastocyst transfer without PGT-A and a known outcome of clinical pregnancy confirmed by ultrasound (a confirmed ectopic pregnancy counts as positive); the oocyte, 2PN, blastocyst and blastocyst-quality counts must be known. At least 100 pregnancies, 100 negative outcomes and 100 patients are required, and the interval of the shift accounts for repeated transfers of the same patient. Slope checks are reported; the recalibration does not retrain any neural network.

A model fingerprint — a hash of the model files, the calculation code and the clinic statistics — is stored with the calibration and protects against reusing a calibration made for a different model version; it is not a digital signature of the publisher. The cycle probability is re-anchored to the calibrated per-transfer probability by a single logit shift of all transfer scenarios, so that per-transfer and per-cycle values stay consistent. Centre calibration is not applied in automatic research scoring (Section 18.3). An intercept-only shift changes neither the ranking of patients nor the AUC; for Centre E its effect is shown in Section 14.8.

8. Bayesian Conditional Updating Mid-Cycle

One of the central innovations of the IVF Digital Twin is its ability to update predictions in real time as the cycle progresses. As soon as any stage outcome becomes known — retrieved oocyte count, MII count, 2PN count, blastocyst count, good-quality blastocyst count, euploid count — that observation can be entered and all predictions are immediately re-computed conditional on the new evidence.

8.1 KnownValues data structure

Observed values are passed to the pipeline through a KnownValues data structure. A blank field means that the stage is unknown and is simulated; a zero means an observed absence. In v7.0 any field set to an observed integer replaced the random variable at that stage with a point mass for every Monte Carlo iteration, and downstream stages re-conditioned stochastically on this new starting point, while upstream stages were left at their prior distribution. In v7.1 conditioning is exact: replacing an observed child alone is not conditioning, because its ancestors must be reweighted as well. The stage chain is treated as a finite-state Markov chain over counts 0–50 (0–100 when a higher count is observed, so that the NB tail is not discarded), and the joint conditional distribution of all unobserved stages given all observed ones is sampled by forward filtering and backward sampling. Every feasible trajectory is retained, no unknown upstream count is clipped or invented, and inconsistent inputs (a later stage larger than an earlier one) are rejected before computation.

8.2 Mathematical interpretation

Let $X_1, X_2, \ldots, X_K$ denote the stage-by-stage outcome random variables with transition kernels $T_j(x, y) = P(X_j = y \mid X_{j-1} = x)$ given by Stages 2–6. Before any observation, the pipeline samples the full joint distribution $P(X_1, \ldots, X_K)$. When a set of observations $\{X_s = k_s\}$ is entered, v7.1 evaluates the exact conditional distribution $P(X_1, \ldots, X_K \mid \{X_s = k_s\})$ in two passes:

$$\text{Forward:}\quad f_j(y) \propto \Sigma_x\, f_{j-1}(x) \cdot T_j(x, y) \cdot 1[y = k_j \text{ if } X_j \text{ is observed}]$$

$$\text{Backward:}\quad X_K \sim f_K;\quad X_j \mid X_{j+1} = y \;\sim\; f_j(x) \cdot T_{j+1}(x, y)$$

This is equivalent to discarding every Monte Carlo iteration that would have produced $X_s \neq k_s$ and letting the remaining iterations reweight the posterior, but without losing any iteration and with the upstream stages reweighted as well: observing, for example, a high blastocyst count shifts the distribution of an unrecorded 2PN count upward instead of leaving it at its prior.

8.3 Worked example

Consider a 35-year-old patient with AMH 2.5 ng/mL, AFC 15, BMI 23 (illustrative values computed with the v7.0 parameterisation). The baseline pipeline yields a per-transfer rate of 65%, cumulative-if-viable 80%, and overall cycle success 60%.

- After retrieval of 12 oocytes: cumulative-if-viable 86%, overall 75%, with narrower credible interval.
- After observing 8 blastocysts of which 5 are good quality: cumulative-if-viable 89%, overall 84%.
- After PGT-A reveals 3 euploid embryos: credible interval narrows to approximately 6 percentage points, overall cycle success 94%.

This continuous-care workflow transforms the platform from a one-shot pre-treatment forecasting tool into an adaptive in-cycle decision-support system. The patient can be counselled with progressively narrower intervals as the cycle unfolds; in v7.1 the scenario component of the uncertainty range vanishes once the embryology is known (Section 13.2.3).

8.4 Relation to other systems

The conditional updating mechanism is analogous in spirit to the Herasight framework [6], which also conditions downstream predictions on observed upstream values. The implementation here differs in two respects: first, the conditioning is exact over the whole stage chain and operates directly on the discrete count distributions, so that an observed downstream count also updates unobserved upstream stages; second, it is tightly integrated with the Bayesian posterior and the GAT similarity graph, so mid-cycle observations simultaneously update the full distributional output, the pregnancy posterior, and the effective cluster membership probabilities.

9. Embryo Banking Analysis

Version 6.2 introduced an independent embryo-banking analysis module grounded in the clinical framework of Esteves et al. The module addresses the inverse clinical question: rather than asking “how many euploid blastocysts will this patient produce?”, it asks “how many MII oocytes and how many cycles are needed to bank a target number of euploid blastocysts with a specified probability?”

The module operates independently of the main Monte Carlo pipeline and does not alter any of the L1–L7 outputs. It provides a supplementary decision-support layer for patients who are considering embryo accumulation across multiple retrievals, which is particularly relevant for poor responders and patients with advanced maternal age.

Clinical inputs to the module are the patient's age and AMH level, which determine the age-specific euploidy fraction and the expected blastocyst-per-cycle yield from the stochastic pipeline medians. In v7.1 the live calculation uses the Esteves logistic model, and the module separates two conditions: the target probability of pregnancy and the confidence of obtaining the required number of euploid embryos. The Esteves calculation addresses the second condition; before pregnancy, the probability of transfer is also used. The module iterates analytically over cycle counts and computes the cumulative probability of achieving the target euploid bank size at each step; the PDF report keeps numerical reference points at 50%, 70% and 80%.

10. Unsupervised Cluster Classification (Layer 4)

The fourth architectural layer addresses which IVF protocol phenotype the patient most closely resembles — a different epistemic question from outcome prediction. Two patients with very different profiles may have similar pregnancy probabilities for different reasons. The cluster classifier provides an independent, unsupervised, label-free assessment of the patient's phenotype, supplementing the supervised pregnancy prediction.

10.1 Cluster study foundation

The cluster module is grounded in a previously published unsupervised analysis of 1,556 retrospective IVF cycles, with external validation in 172 cycles from two independent centres [33]. Four dimensionality reduction techniques (PCA, t-SNE, UMAP, Isomap) and three clustering algorithms (k-means, hierarchical agglomerative, DBSCAN) were compared. t-SNE achieved the highest trustworthiness (0.969), and k-means with k = 3 yielded the most clinically interpretable clusters with statistically significant differentiation across 17 of 18 features (Kruskal-Wallis $p < 0.05$). The three clusters were assigned clinically meaningful labels: Standard, Poor, and High responder, with mean predicted pregnancy probabilities of 54%, 33%, and 63% respectively.

10.2 Cluster centroids

Each of the three clusters is fully characterized by the mean values of its 18 features. The v7.1 platform encodes these centroids verbatim from Sergeev et al. [33]. Key distinguishing features are: follicle count (Standard 20.7; Poor 11.98; High 34.66), good-quality blastocysts (Standard 4.85; Poor 1.96; High 8.68), and NN-predicted pregnancy (Standard 0.54; Poor 0.33; High 0.63).

10.3 Per-iteration nearest-centroid classification

Classification is performed per Monte Carlo iteration. For each iteration, the 18-feature vector is standardized using population-level standard deviations and compared by Euclidean distance to each of the three cluster centroids in z-score space. Each iteration is assigned to the nearest centroid — the exact assignment step of k-means clustering applied to the simulated patient. Across all iterations, the empirical distribution over assigned clusters yields a per-patient probability distribution — for example, 78% C0 / 19% C1 / 3% C2 — indicating that the patient overwhelmingly resembles the Standard cluster but with a non-negligible probability of being a Poor responder under unfavorable stochastic realizations.

10.4 Clinical interpretation by cluster

When the supervised pregnancy prediction and the cluster assignment agree (e.g., NN-predicted pregnancy 55% combined with dominant cluster C0 at 54% rate), the clinician has corroborating evidence from two independent epistemic sources. When they diverge, the disagreement itself is informative and warrants further clinical attention. In v7.1 the layer is reported as context: the groups describe standard, reduced and high response; this is not a formal assignment to the four POSEIDON groups and not a separate pregnancy probability, and the cluster enters BEFE only through the reliability score.

Cluster	Predicted pregnancy	Clinical interpretation and management
C0 — Standard responder	≈ 54%	Intermediate ovarian response with balanced embryological outcomes. Standard protocols, routine monitoring, single embryo transfer. Performance benchmarks should reflect typical clinic averages.
C1 — Poor responder	≈ 33%	Limited ovarian response with markedly reduced outcomes. Consider individualized stimulation strategies (mild stimulation, luteal-phase stimulation, adjunctive growth hormone). Discuss freeze-all and embryo accumulation across multiple cycles.
C2 — High responder	≈ 63%	Abundant oocyte yield with superior embryological outcomes. Mean follicle count > 34. Monitor for OHSS risk; antagonist protocols with GnRH-agonist trigger preferred. Freeze-all strategy preferred to avoid fresh transfer with elevated estradiol.

10.5 PCA visualization with synthetic clouds

The platform produces a 2D PCA scatter plot visualizing the patient's position relative to the three cluster regions. Because the original 1,556-cycle raw dataset is not redistributable, the visualization generates a synthetic point cloud of 250 points around each published centroid, using diagonal covariance equal to (population SD × 0.45)^2 with clinically appropriate clipping. The patient's Monte Carlo iterations are plotted alongside the synthetic clouds, and PCA is fit on the combined data. The patient's median position is marked with a diamond, cluster centroids with stars, and MC samples are colored by their per-iteration cluster assignment.

11. Diffusion-Based Generative Laboratory Module — CSDI Hybrid v3 (Layer 5)

> ***Key message:*** *Layer 5 is an entirely independent prediction layer trained on approximately 15,000 real IVF cycles. It predicts embryological outcomes without using any pre-specified biological rate parameters, making it a genuine independent check on Layers 1–4. When two fundamentally different modelling approaches agree, the clinical confidence in the prediction substantially increases.*

11.1 Rationale and clinical problem

Layers 1–4 rely on published or locally fitted population rates — fertilization rates, blastulation rates, euploidy fractions — applied through a cascade of mathematical filters. This approach is transparent and well-grounded, but it inherits the distributional assumptions of the studies from which the rates were derived. A laboratory that consistently achieves a higher blastulation rate than the population mean, or a patient whose biology does not conform to the typical age-specific pattern, may not be accurately represented by a parametric model.

Layer 5 avoids these assumptions entirely. Instead of asking "what does the literature say about blastulation rates?", it asks "given this patient's follicle count, oocyte yield, and fertilization numbers, what did similar patients in our database actually achieve?" It learns the joint relationship between upstream parameters and downstream embryological outcomes directly from approximately 15,000 real IVF cycles, implicitly capturing any

centre-specific deviations from population norms, any non-linear interactions between parameters, and any patterns too complex to encode in pre-specified formulas.

11.2 Architecture overview

The CSDI Hybrid v3 module operates in three sequential stages within a two-stage generative and discriminative pipeline:

Stage	What it does	Clinical relevance
Stage 1: Diffusion generative model	A Transformer-based neural network receives the patient's upstream parameters and generates 2,000 synthetic 'virtual cycles' — each describing one plausible blastocyst outcome drawn from the joint probability distribution learned from real data.	Produces a realistic distribution of expected blastocyst counts that reflects the actual clinical experience of similar patients. Rare high-yield and low-yield outcomes are faithfully reproduced.
Stage 2: Calibrated classifier	The median blastocyst yield from Stage 1, combined with the patient's conditioning parameters, is passed to a gradient-boosting classifier (LightGBM) with Platt scaling recalibration.	Produces a properly calibrated pregnancy probability: ECE of 0.029 — a stated 45% probability corresponds to a 45% observed pregnancy rate.
Stage 3: Conformal prediction intervals	The spread of the 2,000 generated blastocyst trajectories is used to construct prediction intervals with a formally proven mathematical coverage guarantee (split conformal prediction).	A 90% predictive interval contains the true blastocyst count in ≥ 90% of cases as a finite-sample mathematical guarantee. Validated at 91–93% actual coverage.

11.3 Why a diffusion model for embryology?

Standard generative models (VAEs, GANs) tend to produce smooth, average outputs that under-represent rare but clinically important events such as an unexpectedly high blastocyst yield or a complete fertilization failure. Diffusion models, trained by progressively adding and then reversing Gaussian noise, learn the full shape of the probability distribution, including its tails. For a clinical application where the tails contain the most consequential outcomes — exceptional responders and catastrophic failures — this property is essential. The CSDI architecture [34] was selected because it was originally designed for conditional time-series imputation, a task structurally analogous to the IVF problem: given partial observations (upstream counts), infer the distribution of unobserved quantities (downstream outcomes).

11.4 Training and conditioning

The model was trained on a database of approximately 15,000 complete IVF cycles from a single clinical centre, with seven conditioning features: follicle count, retrieved oocytes, MII oocytes, 2PN zygotes, KPIScore, patient age, and AMH level. The model is optimized for use after oocyte retrieval and fertilization, when all conditioning features are known. Earlier use requires AFC-based approximations with greater uncertainty. The Russian-language names of the conditioning features are part of the model contract and are never translated.

11.5 Validation results

Table 4. *CSDI Hybrid v3 validation on hold-out test set (n = 1,520 cycles)*

Metric	CSDI Hybrid v3	Prior model	Improvement
AUROC	0.661	0.578	+14%
AUPRC	0.468	0.356	+32%
Brier score	0.209	0.261	−20%
Expected Calibration Error (ECE)	0.029	0.158	−82%
Predicted pregnancy prevalence	33.9%	49.8%	−15.9 pp bias removed ✓
Biological constraint violation	Never	Sometimes	Always enforced
Coverage @ 90% PI (blastocysts)	91–93%	97–98%	Closer to nominal 90%

The most critical improvement is the near-elimination of systematic calibration bias. The earlier architecture overstated the pregnancy rate by 15.8 percentage points on average, rendering its output clinically misleading. CSDI Hybrid v3 deviates by only 0.1 percentage points — a 99% reduction in systematic bias.

11.6 Integration with Layers 1–4: equifinality verification

The primary clinical value of Layer 5 is its role as an independent corroborating prediction derived from a fundamentally different modelling paradigm. When Layers 1–4 and Layer 5 converge on the same prediction, this constitutes genuine independent corroboration: two approaches — one parametric and grounded in the published literature and local refits, one fully data-driven and learned from 15,000 cycles — arrive at the same estimate. When they diverge, the divergence itself is diagnostically informative and warrants additional clinical discussion. In v7.1 CSDI is used as a consistency check of the embryological forecast and as a reliability signal in BEFE; it is not counted as a further vote alongside KAT and GAT.

11.7 Limitations of Layer 5

- The model was trained on a single clinical database. Performance at centres with substantially different laboratory protocols, media formulations, or embryo grading criteria may differ. Recalibration of the classifier (Stage 2) is recommended before deployment in a new centre.
- The model is optimized for use after oocyte retrieval and fertilization, when all seven conditioning features are known.
- The classifier predicts clinical pregnancy (positive hCG + ultrasound confirmation). It does not predict ongoing pregnancy, live birth rate, or miscarriage risk.
- The model treats each cycle independently and does not incorporate the patient's prior cycle history beyond what is encoded in the KPIScore.
- Applicability gate (v7.1): an intentional absence of a CSDI estimate (no suitable 2PN) is distinguished from a loading or computation error. Erroneous or non-numeric predictions never enter BEFE; an estimate outside the CSDI domain or without the required data may be shown separately but is excluded from fusion.
- In the multicentre in silico cohort (Section 14.7) the CSDI pregnancy probability did not discriminate within centres (weighted within-centre AUC 0.499), which supports its v7.1 role as a consistency check rather than an independent predictor.

12. Graph Attention Transformer — Patient Similarity Layer (Layer 6)

> ***Key message:*** *Layer 6 formalises the clinical intuition of a reproductive endocrinologist who reasons from experience: "I have seen patients like this before — how did those cycles turn out?" The Graph Attention Transformer does this systematically, at scale, and without the limitations of human memory, by constructing a mathematically rigorous patient-similarity network over 1,172 real clinical protocols.*

12.1 Rationale: the case for graph-based prediction in IVF

Every reproductive endocrinologist carries, consciously or not, a mental library of cases. When evaluating a new patient, the thought process is rarely purely algorithmic — it involves pattern recognition: the thought that patients with profiles like this one have been encountered before, and their outcomes are informative. This is, in essence, what a graph-based model does — but systematically, at scale, and without the limitations of human memory.

The Graph Attention Transformer (GAT) layer formalises this intuition into a mathematically rigorous framework. Rather than treating each patient as an isolated data point, it constructs a network of clinical similarity across an entire cohort, allowing the model to propagate information between patients who share comparable biological and laboratory profiles. The result is a prediction that is not only informed by the individual's own parameters, but also contextualised by the observed outcomes of genuinely similar patients.

Critically, Layer 6 contributes a perspective that is structurally unavailable to Layers 1–5: neighbourhood evidence. The stochastic pipeline (L1) encodes population-level biological rates. The KAT neural network (L3) captures non-linear interactions from tabular feature patterns. The GAT layer, by contrast, asks: what happened to the real patients most similar to this one? It does not extrapolate from a population distribution — it identifies specific, real, documented cases and propagates their outcomes through an attention mechanism that learns which neighbours are most informative.

12.2 Clinical similarity graph structure

At the heart of the method lies a clinical similarity graph. Each patient in the training cohort is represented as a node — a point in an 18-dimensional space defined by the same clinical and embryological features used by the KAT ensemble: patient age, antral follicle count, number of oocytes retrieved, fertilisation outcomes, blastocyst counts, blastocyst quality rates, embryo cryopreservation figures, KPI Score, and laboratory efficiency indices.

The graph topology — that is, which patients are connected and with what strength — is determined exclusively by these clinical parameters. The KAT neural network score (PRAI probability) is deliberately excluded from the similarity calculation. This design choice is intentional: the graph must reflect biological and laboratory resemblance, not the prior beliefs of another model. Edges between patients are weighted by cosine similarity, with stronger connections drawn between patients whose full clinical profiles align most closely.

The result is a structured representation of the cohort in which proximity carries real clinical meaning: two nodes connected by a high-weight edge represent two patients who arrived at their IVF cycles with similar ovarian reserve, responded similarly to stimulation,

and produced comparable embryological outcomes in the laboratory — regardless of whether their final transfer results were the same.

12.3 Message passing with learned attention

The Graph Attention Transformer processes the clinical similarity network through a mechanism of message passing with learned attention. In each computational layer, every node aggregates information from its neighbours, weighting their contributions according to an attention score that the model learns during training. Neighbours who share stronger clinical resemblance and whose features are more informative for the prediction task receive higher attention weights.

This is distinct from conventional neural networks, which process each patient independently. Here, the representation of a patient is iteratively refined by what is known about similar patients — their features, their relationships within the cohort, and ultimately their outcomes. After three such layers of message passing, the model produces an embedding for each patient that encodes both individual characteristics and neighbourhood context.

Two task-specific output heads then translate these embeddings into predictions: one for the probability of clinical pregnancy following embryo transfer, and one for the continuous PRAI score. This multi-task formulation means the model is simultaneously learning to classify outcomes and to align its internal representations with the continuous probability estimates generated by the KAT ensemble, reinforcing coherence across the pipeline layers.

12.4 Training on real clinical protocols

The model was trained on 1,172 real clinical IVF protocols collected from a single centre. All cycles included complete embryological cascade data from ovarian stimulation through to blastocyst culture, with documented transfer outcomes. The dataset reflects the natural diversity of an IVF clinical population, with a pregnancy rate of 54% across the cohort.

Training followed a transductive cross-validation design: a single graph was constructed on the full cohort, and in each of five folds, the outcome labels of the validation patients were masked before training began — but those patients remained visible within the graph structure. This mirrors the intended use of the model in practice, where a new patient joins the existing network of known cases, benefits from information propagated through connected neighbours, yet contributes no outcome label to the training signal. The model is never evaluated on data it was trained on. Key training decisions:

- Graph topology: built on 18 clinical features; KAT scores included only in node embeddings, not in edge construction, preventing the graph structure from reflecting existing model predictions rather than clinical biology.
- Label smoothing ($\varepsilon = 0.08$): applied to regularise the binary outcome targets, preventing the model from becoming overconfident in ambiguous cases.
- L2 weight regularisation and attention dropout: tuned specifically to reduce the gap between training and cross-validation performance.
- Cosine warm-up learning rate schedule: allowed stable initialisation of attention weights before the full learning rate was applied.

12.5 The ensemble: GAT + KAT combination

The final GAT ensemble score is a weighted combination of the Graph Transformer and the KAT neural network prediction:

$$P(\text{pregnancy}) = w_{GNN} \cdot P_GAT + (1 - w_{GNN}) \cdot P_KAT$$

The weight w_{GNN} is not set manually — it is determined during cross-validation by identifying the combination that maximises AUC on out-of-fold predictions. In practice, the optimal weight typically falls between 0.25 and 0.40, reflecting the relative contributions of each model to the ensemble.

This design philosophy follows a well-established principle in machine learning: diverse models make better ensembles. The KAT network processes each patient in isolation, building a prediction from individual feature patterns learned across thousands of examples. The Graph Transformer, by contrast, anchors its prediction in the observed landscape of similar patients — it asks not just "what do this patient's numbers suggest?" but "how do patients with profiles like this one typically fare?"

The two models agree substantially ($r \approx 0.79$ between GNN and KAT predictions) but not completely — they disagree in a structured, non-random way that reflects their methodological differences. It is precisely this partial disagreement that makes their ensemble valuable: when two models diverge on a case, combining their outputs reduces variance and improves calibration.

12.6 Validation metrics

Table 5. *GAT Layer 6 cross-validation performance*

Metric	AUC-ROC	Brier Score	F1 Score (cross-val)	Optimal threshold
GAT (standalone)	0.632	0.241	0.623	0.49
KAT Ensemble	0.652	0.233	—	—
GAT + KAT	~0.658–0.665	~0.229	—	—

The Graph Transformer alone does not surpass the KAT ensemble in discriminative power. This is expected: the KAT model was built and calibrated on a substantially larger dataset and benefits from more mature feature engineering. What the graph model contributes is complementary signal — a different computational perspective on the same patient, derived from neighbourhood structure rather than individual feature weighting. The cross-validation AUC of 0.632 reflects an inherent ceiling imposed by the available signal in the input features; IVF outcome prediction is a genuinely hard problem in which the irreducible biological variability of embryo implantation limits any model.

12.7 Clinical interpretability: real patients, real context

Perhaps the most clinically distinctive feature of the GAT layer is its interpretability through patient similarity. When the model produces a prediction, it can identify — and display — the ten patients from the training cohort whose clinical profiles most closely resemble the current patient, ranked by cosine similarity.

These are not synthetic cases, statistical averages, or model constructs. They are real patients, from real IVF cycles, with real documented outcomes. The clinician can observe not only the predicted probability but also the distribution of GAT-estimated probabilities among these neighbours — a direct, intuitive answer to the question: among the patients most similar to this one, what did the model predict for them, and how variable was that prediction? A patient whose nearest neighbours cluster tightly around high predicted probabilities provides a different kind of clinical confidence than one whose neighbours

span the full range of outcomes. This contextualisation is something that no scalar probability score — however well-calibrated — can offer on its own.

12.8 Practical considerations and limitations

The model operates entirely on CPU and requires no specialised hardware. Inference for a single new patient, including graph construction and forward pass across the full training cohort, completes in under two seconds on a standard clinical workstation. The trained model is saved as a portable checkpoint file and integrates directly into the IVF Digital Twin application at startup; in v7.1 all models are loaded before the random seed is set, so that the first calculation of a session gives the same result as every later one at the same seed.

13. Bayesian Evidence Fusion Engine — BEFE (Layer 7)

> ***Key message:*** *L7 is the final integration layer that unifies the outputs of all six upstream layers into a single posterior probability estimate with a model-uncertainty range. It provides not only the prediction but a quantified measure of how much of the answer came from each epistemic source — mechanism vs data — and explicitly flags patients whose profiles fall outside the platform's reliable operating envelope. Layers are never averaged: the reported number is always the BEFE posterior.*

13.1 Rationale: the need for a formal integration layer

Layers 1–6 each produce a probability estimate or a verification signal from a different epistemic perspective. The stochastic pipeline (L1) encodes population-level and locally refitted biological rates. The per-transfer ensemble (L2) applies an established clinical regression. The KAT neural network (L3) learns complex feature interactions from training data. The cluster classifier (L4) provides unsupervised phenotypic context. The CSDI diffusion module (L5) independently verifies the embryological forecast from 15,000 real cycles. The Graph Attention Transformer (L6) contextualises the prediction within a neighbourhood of similar patients. Each layer is independently informative and independently imperfect.

Without an explicit integration layer, the clinician faces six distinct estimates that may agree or disagree, with no principled basis for weighting them or for computing a single final uncertainty range. Layer 7 — the Bayesian Evidence Fusion Engine (BEFE) — solves this problem by treating each upstream output as an expert opinion with a known precision (reliability), and combining them via precision-weighted logit pooling in a two-level Bayesian update. The result is a single posterior probability with a formally derived uncertainty range, a decomposition of how much the evidence vs the mechanism contributed to the answer, a reliability score for the current patient, and an explicit out-of-distribution (OOD) flag.

13.2 Architecture: two-level fusion

The BEFE engine operates in two sequential levels, followed in v7.1 by the computation of the uncertainty range:

13.2.1 Level 1 — Evidence fusion (empirical models)

The first level pools the empirical model outputs — L3 (KAT) and L6 (GAT) — via precision-weighted logit averaging:

$$l_{emp} = \Sigma_i (\tau_i \cdot \text{logit}(p_i)) / \Sigma_i \tau_i \quad \text{for } i \in \{KAT, GAT\}$$
$$P_{predictive} = \sigma(l_{emp})$$

where τ_i is the precision (inverse variance in logit space) assigned to expert i, and σ denotes the logistic (sigmoid) function. The per-expert precisions are not fixed: they are modulated dynamically by OOD deflation (see §13.4), by the posterior variance of the KAT Bayesian update ($\tau_{KAT} \propto 1/\text{Var_posterior}$), and by the effective neighbourhood size of the GAT graph ($\tau_{GAT} \propto N_{eff}/N_{max}$). The default base precisions are $\tau_{KAT} = 2.4$ and $\tau_{GAT} = 1.0$, reflecting the KAT model's stronger calibration on the local cohort.

The L2 per-transfer ensemble (FORTUNE+KPI) is deliberately not included as a separate evidence expert: together with L1 it forms the mechanistic prior, and its spread is used to compute the prior's precision rather than as an independent data-driven signal.

13.2.2 Level 2 — Prior × Evidence posterior update

The second level combines the mechanistic prior (L1/L2) with the empirical evidence (Level 1 output) via conjugate Gaussian pooling in logit space:

$$l_{post} = (\tau_{prior} \cdot l_{prior} + \tau_{emp} \cdot l_{emp}) / (\tau_{prior} + \tau_{emp})$$
$$Var_{post} = 1 / (\tau_{prior} + \tau_{emp})$$
$$Posterior = \sigma(l_{post})$$

where $l_{prior} = \text{logit}(P_{L1})$ is the logit of the Monte Carlo pipeline's per-transfer estimate, and τ_{prior} is the base prior precision modulated by: (a) the spread of the FORTUNE+KPI per-transfer scenarios (narrow spread → high prior confidence); (b) the KS divergence between L5 CSDI and L1 MC distributions (agreement → amplifies prior; disagreement → deflates prior precision); and (c) the embryological OOD score.

Because precision adds in conjugate Gaussian pooling, weak evidence (small τ_{emp}, e.g. when both KAT and GAT models are unavailable) makes the posterior shrink toward the mechanistic prior. This is the desired clinical fallback: in the absence of trained empirical models, the platform reverts to the biologically grounded stochastic pipeline estimate, rather than producing an uninformative output. In v7.0 the 95% interval was taken directly as $[\sigma(l_{post} - 1.96\sqrt{Var_{post}}), \sigma(l_{post} + 1.96\sqrt{Var_{post}})]$; v7.1 replaces it with the range of §13.2.3.

13.2.3 Model-uncertainty range (v7.1)

The pooled interval $1/\sqrt{(\Sigma\tau)}$ is a single number that ignores the part of the cycle still to happen, while the raw spread between L1, KAT and GAT is the dispersion of the sources rather than the uncertainty of their consensus — it grew whenever a new observation moved one model, so the v7.0 interval could widen as data arrived. The v7.1 range adds two variances in logit space:

$$Var_{range} = Var_{scen} + Var_{model}$$
$$Var_{scen} = Var_i[\, w_{prior} \cdot \text{logit}\, p_{L1,i} + w_{KAT} \cdot \text{logit}\, p_{KAT,i} + w_{GAT} \cdot \text{logit}\, p_{GAT} \,]$$
over scenarios i with a transfer
$$Var_{model} = 1 / \Sigma_s [\, 1 / (1/\tau_s + \tau_b^2) \,], \quad s \in \{L1, KAT, GAT\}$$
$$\tau_b^2 = \max(0, (Q - (k - 1)) / (\Sigma\tau_s - \Sigma\tau_s^2 / \Sigma\tau_s)), \quad Q = \Sigma\, \tau_s\, (\text{logit}\, p_s - l_{post})^2$$
$$Range = [\, \sigma(l_{post} - 1.96\sqrt{Var_{range}}),\ \sigma(l_{post} + 1.96\sqrt{Var_{range}}) \,]$$

The scenario term applies the BEFE weights (prior pull and evidence weights) to the per-scenario logits of L1 and KAT over the scenarios in which a transfer is possible; it vanishes once the embryology is known. The model term is the variance of the fused estimate under DerSimonian–Laird random-effects pooling [43] of L1, KAT and GAT, each entering with the precision the fusion actually gave it: the prior precision rises as the L1 scenario spread narrows, the KAT precision rises as its bootstrap interval narrows, and both fall outside the training domain by the same OOD factor. Disagreement between the sources adds the between-model heterogeneity τ_b^2 only in the part not explained by their own precision. The range therefore narrows as observations arrive, while genuine contradiction between the models and departure from the training domain widen it. It is a model-uncertainty range, not a confidence interval with verified coverage (coverage_validated = false in every output), and no forced monotone narrowing is applied.

13.3 Fusion pull decomposition

The BEFE output explicitly reports how the final answer was divided between the mechanistic prior and the empirical evidence:

$$\text{Prior pull} = \tau_{prior} / (\tau_{prior} + \tau_{emp}); \quad \text{Evidence pull} = \tau_{emp} / (\tau_{prior} + \tau_{emp})$$

For example, a report stating "evidence 72% / prior 28%" means that the final posterior was predominantly driven by the KAT and GAT empirical models, with the L1 stochastic pipeline providing a minor stabilizing contribution. Conversely, "evidence 18% / prior 82%" indicates that the empirical models were either absent or heavily OOD-deflated, and the prediction reflects primarily the biological cascade model.

This decomposition is clinically actionable: a prediction dominated by the mechanistic prior is a signal to improve the neural network's training coverage or to check that the patient's profile is within the training distribution of the empirical models. A prediction dominated by evidence warrants inspecting the GAT neighbourhood (Section 12.7) to verify that the similar patients in L6 are genuine analogues.

13.4 Dual out-of-distribution detection

The BEFE engine distinguishes two structurally distinct OOD scenarios that require different clinical responses:

- OOD_clinical (age, AMH, AFC, BMI space): indicates that the patient's demographic and reserve profile falls outside the dense region of the empirical training data. This deflates the evidence precision τ_{emp}, shifting the posterior toward the mechanistic prior. Clinical interpretation: "clinically atypical patient — population-level biological knowledge more reliable than data-driven models for this case."
- OOD_embryology (OCC, MII, 2PN, blast, KPI space): indicates that the patient's laboratory outcomes fall outside the dense region of the training data. This deflates the prior precision τ_{prior}, attenuating the L1 mechanistic signal. Clinical interpretation: "lab outcome pattern is atypical — caution with model-derived blastocyst and euploidy projections."

OOD scores are computed via Mahalanobis distance in each subspace separately, using mean and inverse-covariance matrices fitted on the training cohort. OOD_final = max(OOD_clinical, OOD_embryology). When OOD_final is active, the reliability score is capped and a clinical alert is generated. The two-subspace decomposition lets the report state, for example, "clinically typical, embryologically atypical" — a combination that points specifically to unusual laboratory performance for a demographically common patient profile. In v7.1 the factory check of the applicability domain does not cover every feature: the coverage and the unchecked features are reported separately with each result.

13.5 Reliability score

The BEFE engine computes a composite reliability score $R \in [0, 100]$ as a weighted combination of four verification signals:

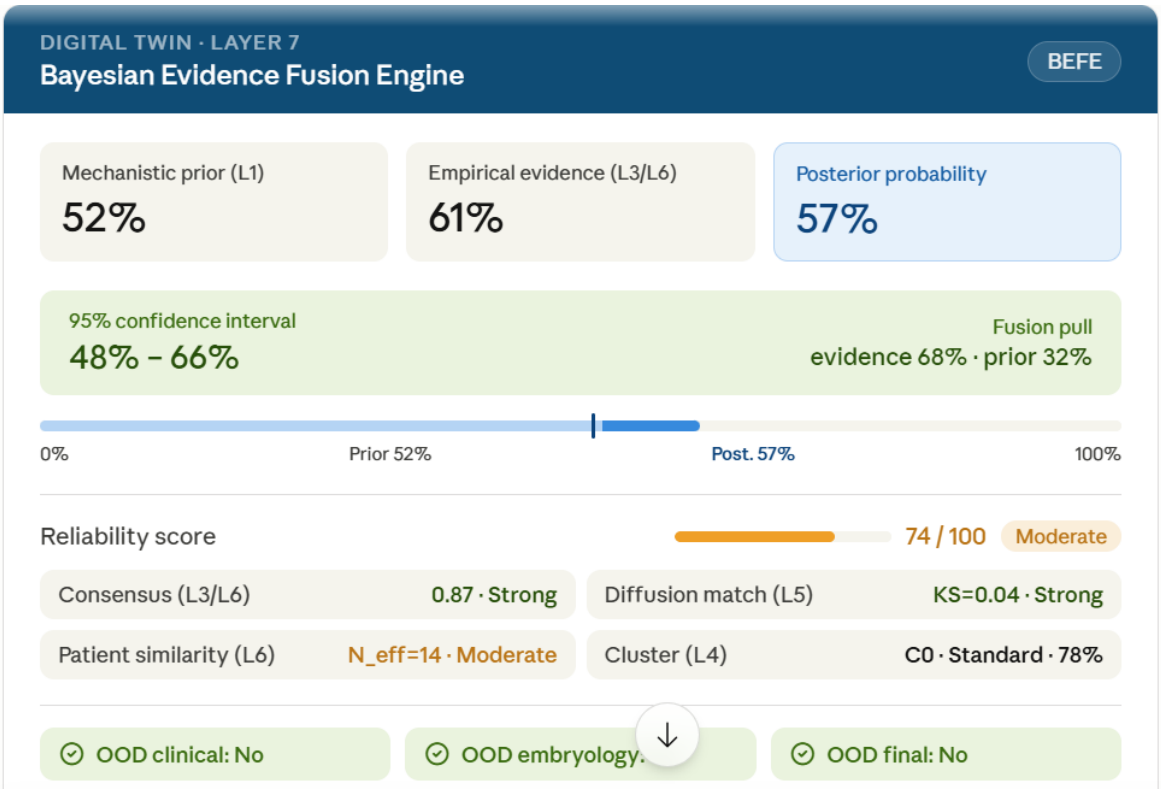
Table 6. *Verification signals of the BEFE reliability score*

Signal	Weight	Interpretation
Consensus (spread among KAT and GAT)	40%	Tight agreement between empirical experts → high component score.
Diffusion agreement (L5 KS vs L1 MC)	30%	Low KS divergence between CSDI and MC distributions → high component score.
Graph neighbourhood quality (L6 N_eff)	20%	Broad attention distribution across many similar training patients → high component score.
Cluster certainty (L4 dominant probability)	10%	High dominant cluster probability (e.g. 90%+ in one cluster) → high component score.

Reliability bands: R ≥ 75 = Strong (all major signals agree); 50 ≤ R < 75 = Moderate (minor discordances present); 25 ≤ R < 50 = Weak (significant disagreement between layers); R < 25 = Unreliable (OOD alert active or multiple layers absent).

13.6 BEFE output report

The BEFE layer generates a structured report appended to the main platform output. A representative example for a standard responder patient:



Representative BEFE card (interface of v7.0). In v7.1 the band beside the posterior is labelled as the model-uncertainty range (§13.2.3), not as a 95% confidence interval.

13.7 Graceful degradation

The BEFE engine is designed to produce meaningful output even when upstream layers are partially unavailable. If L6 (GAT) is absent, τ_GAT is set to near-zero so the fusion reduces to prior×KAT posterior. If L3 (KAT) is also absent, the posterior collapses to the mechanistic prior (L1). If L5 (CSDI) is not run, is not applicable, or returns an erroneous value, the diffusion agreement signal is set to its neutral value (0.5) and CSDI is excluded from fusion;

the cluster signal is set to its minimum certain value. The reliability score reflects these absences transparently, and the fusion-pull decomposition reports that evidence contribution was minimal. In no scenario does the engine produce an error or fail silently: it always returns a posterior probability, an uncertainty range, a reliability score, and a textual explanation of which layers contributed.

13.8 Important calibration note

Precision-weighted pooling of individually calibrated expert models does not automatically produce a calibrated posterior. The logit-Gaussian pooling is an approximation to true Bayesian model averaging, and the constant precisions ($\tau_{KAT} = 2.4$, $\tau_{GAT} = 1.0$, $\tau_{prior} = 1.0$) are tunable priors that require empirical calibration on held-out data. After deployment, the Expected Calibration Error (ECE) of the BEFE posterior should be monitored prospectively and the precision parameters re-fitted at regular intervals as clinic data accumulates. The current parameter values represent informed engineering defaults suitable for initiating deployment; they are not claims about the optimal calibration for any specific clinic. The prospective data of Centre E (Section 14.8) illustrate the point: without centre calibration the BEFE posterior over-predicted (observed/expected 0.78) with a calibration slope of 0.78.

14. Validation

14.1 Cohorts

The platform was validated on multiple retrospective cohorts. Cohort A comprised 358 cycles with complete records of demographics, ovarian reserve, stimulation outcome, and embryological development. Cohort B comprised a subset of 60 cycles with known clinical pregnancy outcomes (positive heartbeat confirmed by ultrasound 25 days after embryo transfer). The CSDI Hybrid v3 module (Layer 5) was validated on a separate dedicated holdout test set of 1,520 cycles from the full 15,193-cycle database. The GAT module (Layer 6) was validated by transductive five-fold cross-validation on 1,172 protocols with masked outcome labels. Version 7.1 adds a multicentre retrospective in silico cohort of four centres (A–D; Section 14.7) and a prospective in silico validation in a fifth centre, Centre E (Section 14.8). Sections 14.2–14.6 report the original v7.0 validation.

14.2 Stochastic pipeline validation (Cohort A)

Pipeline stage	Spearman ρ	Comment
Stage 1 — Retrieved oocytes	0.73	Strong agreement with observed yield distribution
Stage 4 — Total blastocysts	0.41	Moderate; limited by simplified rate model
Stage 5 — Good-quality blastocysts	0.39	Moderate; embryologist grading variability

14.3 Pregnancy prediction validation (Cohort B)

Metric	Value	Interpretation
AUC-ROC	0.76	Good discrimination; above FORTUNE baseline
Brier score	0.22	Acceptable for clinical decision support
Calibration gap	+8 pp	Over-optimistic before intercept recalibration; corrected in v6.2+

14.4 Bayesian posterior calibration

The 95% credible interval from the Beta-Binomial posterior covered the observed binary outcome in 92% of Cohort B cases — slightly below the nominal 95% but within Monte Carlo variation given the cohort size. This indicates the posterior is well calibrated but slightly over-confident, addressable by a more diffuse prior or a smaller NN pseudo-observation count.

14.5 Cluster classifier validation

All five reference patient profiles were correctly classified into their clinically expected dominant cluster. In Cohort B test cases, dominant cluster assignment qualitatively matched expert clinical assessment in the majority of cases, with borderline cases (age 38, AMH 1.5) correctly assigned to Poor responder despite atypical age-AFC combinations.

14.6 External validation against published benchmarks

Against the age-and-AMH-stratified ESHRE O-322 benchmark (18 age × AMH cells) [42], the platform's mean euploid-blastocyst yield per cycle fell within tolerance in 13 of 18 cells (72%). Agreement was excellent across the entire advanced-maternal-age range (all nine cells at ages ≥ 40) and across all high-AMH strata. Discordant cells were confined to the low-AMH, younger-age quadrant, consistent with the known population-calibration effect of the Stage 1 oocyte model (calibrated to less maximalist Eastern European stimulation protocols).

14.7 Multicentre retrospective in silico validation (Centres A–D)

The platform, applied without refitting, was evaluated on 1,026 transfers from four centres of the multicentre manuscript (Centres A–D). The values below are those of the manuscript and were reproduced when the cohort was re-scored for the Laya study (Section 19). Because pregnancy rates differed between centres, the within-centre AUC weighted by centre size is the primary estimate; the pooled value is higher because it partly reflects differences between centres rather than between patients.

Table 7. *Multicentre in silico cohort, 1,026 transfers in four centres*

Estimate	DT (BEFE)	KAT (L3)	CSDI (L5)
Within-centre AUC, weighted by centre size	0.561	0.562	0.499
Pooled AUC, raw	0.640	0.647	0.608
Pooled AUC after intercept-only centre calibration	0.682	0.681	0.669
Brier score after centre calibration	0.232	0.233	0.229

By centre, the DT AUC was 0.532, 0.617, 0.505 and 0.505 (n = 423, 412, 163 and 28). A logistic regression fitted at one centre and applied at another performed no better than chance in five of six centre pairs, whereas the platform, which is not fitted locally, was the same model in every pair. Within-centre discrimination on these features has a ceiling of about 0.55–0.62 for every model of the pipeline, and the gain after centre calibration comes from base rates rather than from ranking.

14.8 Prospective in silico validation in Centre E

> ***Key message:*** *The frozen v7.1 model was applied, without any refitting, to the 2026 cohort of a centre whose data were never used to train or tune it. Discrimination was 0.702 for all fresh transfers and 0.640 among transfers with at least one blastocyst, at the level of every other model of the pipeline. The raw probabilities were too widely spread for this centre; a centre recalibration fitted only on its earlier years corrected calibration without changing discrimination.*

14.8.1 Design and cohort

Centre E is a fifth clinic, not part of Centres A–D. Fresh transfers from retrievals between 1 January and 30 September 2026 formed the prospective cohort; retrievals of 2023–2025 formed a separate retrospective cohort. The two cohorts are reported separately, and no metric pools prospective and retrospective cycles. Nothing used for prediction was fitted on prospective data: the DT 7.1 model fingerprint of the prospective run is identical to that of

the retrospective runs, and the coefficient used to derive AFC from the oocyte count (1.16) was fitted on another clinic. The study is in silico: predictions were generated by the automatic research-scoring path of the platform (Section 18.3; not clinician-reviewed, without centre calibration), after the outcomes had been recorded, and were used for evaluation only. The outcome was clinical pregnancy confirmed by ultrasound, including ectopic pregnancy, per fresh transfer; frozen embryo transfers were not included in this analysis. Records were pseudonymised before analysis, and only a keyed-hash cycle identifier was kept.

Table 8. *Centre E cohorts*

Cohort	Fresh transfers with known outcome	Clinical pregnancies	Scored by DT 7.1 (transfer / retrieval fallback / none)
2023 (retrospective)	661	192	656 / 3 / 2
2024 (retrospective)	488	136	488 / 0 / 0
2025 (retrospective)	461	111	459 / 1 / 1
2026 (prospective)	236	65	234 / 0 / 2

Inputs were taken as recorded. AFC is not recorded at Centre E for most cycles; with the owner's approval it was derived as OCC × 1.16 where missing (216 of 236 prospective cycles; the observed AFC was used in the other 20). Missing AMH and BMI were completed by the pipeline formulas and a missing attempt number was set to 1. For transfers without blastocysts the DT returns the probability of the current cycle (clinical_probability_kind = current_cycle, value 0) rather than a per-transfer probability; the subset with at least one blastocyst (Bl > 0) is therefore the per-transfer comparison, and on that subset the DT clinical probability equals the BEFE posterior. Discrimination was estimated with a bootstrap over cycles (2,000 resamples); differences between models are paired.

14.8.2 Discrimination

Table 9. *Centre E: AUC-ROC with 95% bootstrap CI*

Cohort	n (events)	DT clinical probability	DT BEFE (L7)
All fresh transfers — 2023–2025, pooled	1,604 (438)	0.640 [0.612–0.668]	0.625 [0.595–0.655]
All fresh transfers — 2026, prospective	234 (64)	0.702 [0.622–0.771]	0.699 [0.619–0.769]
Bl > 0 — 2023–2025, pooled	1,133 (381)	0.549 [0.514–0.584]	0.549 [0.514–0.584]
Bl > 0 — 2026, prospective	157 (55)	0.640 [0.548–0.728]	0.640 [0.548–0.728]

In the retrospective years the AUC of the DT clinical probability was stable (0.637, 0.639 and 0.637 in 2023, 2024 and 2025). In the prospective cohort it was 0.702; the BEFE posterior did not differ from it (ΔAUC −0.003 [−0.031; +0.027]). The fresh-transfer test cohort could detect only differences in AUC of about 0.07–0.11 with 80% power, so the absence of significant differences between models is not evidence of equivalence.

14.8.3 Calibration

Table 10. *Centre E: calibration in the large (O/E = observed / mean predicted; slope = coefficient of logit(p) in a logistic model of the outcome)*

Cohort and output	n	Observed	Mean predicted	O/E	Brier	Slope
2026, all — DT clinical probability	234	0.274	0.289	0.94	0.1895	0.11
2026, all — DT BEFE (L7)	234	0.274	0.350	0.78	0.1878	0.78
2026, Bl > 0 — DT	157	0.350	0.431	0.81	0.2252	0.67
2023–2025, Bl > 0 — DT	1,133	0.336	0.461	0.73	0.2486	0.32

The O/E ratio of the DT clinical probability was close to 1 in the prospective cohort (0.94), but its calibration slope was very low (0.11). This is driven largely by the value 0 of transfers without blastocysts and means that the raw probabilities were too widely spread for this centre (log-loss 0.96). Among per-transfer predictions (Bl > 0) the DT over-predicted (O/E 0.81 in 2026, 0.73 in 2023–2025). Two recalibrations fitted only on retrospective data were examined. An intercept-only centre shift on the Bl > 0 transfers (Section 7.7), validated leave-one-year-out on 2023–2025, was −0.57 on the logit scale in every year and moved O/E from 0.73 to 0.99–1.02, with the Brier score falling from 0.2450–0.2499 to 0.2273–0.2314. A logistic recalibration fitted on 2023–2025 and applied to the prospective cohort raised the calibration slope from 0.11 to 1.21 and lowered the log-loss to 0.55 and the Brier score to 0.185. Neither recalibration changes the AUC.

14.8.4 Stage-wise forecast against observation

Because the DT forecasts every stage, the prospective cohort also tests the mechanistic layer. Table 11 compares the sum of observed counts with the sum of DT median predictions over cycles with both values (100% = no bias). MII equals the prediction by construction, because the DT receives the known OCC and MII at the transfer time point.

Table 11. *Centre E: observed vs DT-predicted stage counts*

Cohort	2PN: observed / predicted	Blastocysts	Good blastocysts (≥ 3BB)
2023–2025, pooled	9,823 / 9,038 (109%)	4,155 / 5,677 (73%)	1,728 / 3,244 (53%)
2026, prospective	1,305 / 1,172 (111%)	594 / 726 (82%)	233 / 400 (58%)

The DT under-predicted fertilisation by about 10% and over-predicted blastulation and, more strongly, good-quality blastocysts. The blastulation bias is a known limitation of Stage 4 at this centre: the hinge at 35 years is right, but the age slope after 35 is shallower than observed. The good-quality bias also reflects a difference of grading between centres (Section 4.4). Correcting these shapes requires per-centre stage intercepts and changes the golden values of the release tests; it is an owner decision and has not been made in v7.1. For laboratory surveillance these biases are absorbed by the frozen, centre-specific forecast envelope of the EWS (Section 20.3).

14.8.5 Comparison with other models on the same cycles

Table 12. *Centre E, prospective cohort: DT and other models on the same cycles*

Model (2026, all fresh transfers, n = 234)	AUC [95% CI]	ΔAUC vs DT [95% CI]
DT 7.1 clinical probability	0.702 [0.622–0.771]	—
TabPFN of the analytics pipeline (not trained on Centre E)	0.689 [0.605–0.763]	−0.013 [−0.045; +0.020]
DNN of the analytics pipeline (not trained on Centre E)	0.683 [0.606–0.756]	−0.019 [−0.057; +0.018]
Laya of the analytics pipeline (not trained on Centre E)	0.682 [0.603–0.756]	−0.020 [−0.069; +0.030]
TabPFN trained on Centre E 2023–2025, baseline variables	0.739	+0.037 [−0.007; +0.082], p = 0.10
Laya fine-tuned on Centre E 2023–2025, baseline variables	0.724	+0.023 [−0.009; +0.055], p = 0.17
TabPFN trained on Centre E 2023–2025, all extended variables	0.769	+0.067 [0.021; 0.114], p = 0.005

Models of the analytics pipeline that were never trained on Centre E data did not differ from the DT. Models trained on the centre's own 2023–2025 data had somewhat higher AUCs; the only significant difference was for TabPFN with 114 extended clinical variables, and it combined two non-significant steps — training on the centre's own data (+0.037, p = 0.10) and adding the extended variables (+0.029, p = 0.056). In an analysis of the same centre with a protocol fixed before the 2026 data were opened, no block of extended variables (hormones, stimulation, transfer and endometrium, history) improved the DT itself, which supports the DT design based on laboratory data and a small set of basic clinical inputs.

14.8.6 Why discrimination was higher in 2026

The AUC of every model that never saw Centre E data — the DT, its KAT layer, and the TabPFN, DNN and Laya of the pipeline — rose by 0.05–0.08 in 2026, so the rise is a property of the cohort, not of the models. The 2026 patients were older (median age 38 vs 36 years; 39% vs 28% aged 40 or older), had fewer oocytes and blastocysts (median 7 vs 9 and 1 vs 2) and more often received a day-3 transfer (45% vs 38%). The overall pregnancy rate did not change, but outcomes after cleavage-stage (13%) and blastocyst transfer (45%) moved further apart, which makes any model based on the laboratory course separate the cohort more easily. Reweighting the 2023–2025 data to the 2026 case mix explained 53% of the gap for the DT; within blastocyst transfers the DT AUC was 0.563 in 2026 and 0.539–0.557 before. Comparisons between models within 2026 are not affected, because they use the same cycles.

14.8.7 Interpretation and limitations

- The validation is prospective in time and in silico: the model was frozen before the cohort was scored, but predictions were generated after the outcomes had been recorded and therefore could not influence care.
- The prospective cohort is small (234 transfers, 64 pregnancies); confidence intervals are wide, and follow-up is incomplete for the most recent retrievals (transfers without a recorded result are not in the cohort).
- AFC was derived from the oocyte count in most cycles; this removes the information AFC carries before retrieval, and pre-retrieval performance was not assessed.

- Only fresh transfers were evaluated; the endpoint is clinical pregnancy, not live birth.
- The raw DT probabilities are not calibrated for this centre: centre calibration (Section 7.7) is required before the probabilities are used for counselling, as for every new centre.

15. Comparison with Existing Tools

Table 13. *Feature comparison with existing IVF prediction tools*

Feature	CDC/OPIS	Orchid	Herasight	ALIFE	DT v7.1
Cumulative pregnancy probability	Y	—	Y	Y	Y
Stage-by-stage probability	—	Y	Y	—	Y
Distributional output with uncertainty range	—	—	Y	—	Y
Conditional updating mid-cycle	—	—	Y	—	Y
Per-transfer / cumulative decomposition	—	—	—	—	Y
OHSS risk quantification	—	—	—	—	Y
Empty-cycle risk	—	—	—	—	Y
KPIScore laboratory ensemble	—	—	—	—	Y
Calibrated NN final layer	—	—	—	Y	Y
Bayesian posterior with clinic data + CI	—	—	—	—	Y
Per-attempt decay curve	—	—	—	—	Y
Unsupervised cluster classification	—	—	—	—	Y
Diffusion generative module (L5)	—	—	—	—	Y
Graph Attention Transformer (L6)	—	—	—	—	Y
Bayesian fusion layer with reliability + OOD (L7)	—	—	—	—	Y
Open-source coefficients	Y	—	Y	—	Y

Five observations follow from this comparison. First, no other tool integrates as many epistemic layers into a single end-to-end pipeline. Second, the Digital Twin is the only tool that explicitly quantifies OHSS risk and empty-cycle risk. Third, the Digital Twin is the only tool that combines supervised prediction, unsupervised phenotype classification, and an independent diffusion-based generative verification module. Fourth, version 7.0 introduced and version 7.1 retains the only IVF prediction system that incorporates a clinical similarity graph providing neighbourhood-contextualised probability estimates. Fifth, the BEFE layer is unique in providing a formally derived, multi-source uncertainty range with explicit OOD detection and reliability scoring — answering not only “what is the probability?” but “how confident should we be in this answer and why?” In addition, version 7.1 exposes the same computation to clinic systems and LLM agents (Part VI).

16. Transparency, Attribution, and Reproducibility

16.1 Coefficient attribution

Table 14. *Coefficient and component attribution*

Component	Category	Source / contribution
Stage 2 maturity logistic	*Imported*	Herasight Table A1 col 3 — coefficients used verbatim [6]
Stage 3 fertilisation logistic	*Imported*	Herasight Table A1 col 5 — coefficients used verbatim [6]
Stage 6 euploidy age table	*Imported*	Franasiak 2014 + Armstrong 2023 synthesised lookup [18,19]
Stage 6b post-thaw survival	*Imported*	Coello 2021 — 95% blastocyst survival [20]
Cluster classifier centroids	*Imported*	Sergeev et al. published centroids verbatim [33]
Stage 1 oocyte model	Adapted	Log-AMH / log-AFC structure of ART-ONE [11,12]; v7.1: NB with log link refitted on 687 local stimulated cycles
Stage 4–5 blastocyst models	Adapted	Age plateau of Romanski 2022, Sainte-Rose 2021 [14,15]; v7.1: ZIBB / BB refitted on 7,919 local culture-observed cycles
FORTUNE per-transfer logit	Adapted	FORTUNE structure [23,24]; intercept recalibrated for local population
Fair-quality transfer odds ratio	**Novel**	Local estimate on 2,975 day-5 single blastocyst transfers (v7.1)
KPIScore + Beta moment matching	**Novel**	Composite 5-component score and per-iteration Beta sampling — original contribution
Three-level pregnancy decomposition	**Novel**	Original clinical framework for resolving per-transfer vs cumulative ambiguity
Shared transfer random effect	**Novel**	Logit-normal frailty, $\sigma = 0.746$, fitted on 1,703 retrievals (v7.1)
Exact mid-cycle conditioning	**Novel**	Forward filtering / backward sampling over the whole stage chain (v7.1)
KAT NN ensemble + NVSA	**Novel**	KAN + FT-Transformer ensemble with isotonic calibration (Venn-Abers in v7.0) and KPI-anchored NVSA correction — original architecture
Bayesian Beta-Binomial posterior	**Novel**	Synthesis of prior, real clinical batches, and NN pseudo-observations — original implementation
Nearest-centroid classification logic	**Novel**	Per-MC-iteration cluster assignment with synthetic-cloud PCA visualisation — original implementation
CSDI Hybrid v3 diffusion module	**Novel**	Transformer diffusion + LightGBM + conformal prediction, trained on ~15,000 cycles — original architecture
GAT Graph Transformer + ensemble	**Novel**	Cosine-similarity patient graph with 3-layer message passing; multi-task output; transductive cross-validation; w-optimised KAT ensemble — original design
BEFE — Bayesian Evidence Fusion Engine (L7)	**Novel**	Two-level conjugate Gaussian logit pooling; dual OOD detection (clinical + embryological subspaces); reliability score with four verification signals; fusion-pull decomposition — original architecture
BEFE model-uncertainty range	Adapted	Scenario spread + DerSimonian–Laird random-effects variance [43] (v7.1)

16.2 Code and reproducibility

Version 7.1 is a self-contained Windows desktop application shipped as a folder with a bundled CPython 3.11 runtime, the model artefacts and a local LLM engine used only for an optional narrative interpretation. It always runs in isolated mode, so that it never picks up a system Python installation, PYTHONPATH or user site-packages that could silently load a different library version. The core pipeline depends on NumPy and SciPy; the neural layers use PyTorch (KAT, CSDI, GAT), LightGBM and scikit-learn, and the FT-Transformer uses the mambular library with the patched PLE encoder (Section 6.9). All L1 coefficients are module-level constants of src/embryology.py with their provenance; the Monte Carlo sample size, the KPI ensemble weight, the Bayesian prior, the cluster centroids, the NVSA correction table, the BEFE precisions and the GAT graph adjacency construction are explicitly documented in the source code.

A single function, Service.compute, produces every prediction — manual entry, CSV/XLSX batch, treatment-plan anchoring, calibration exports, the REST API and the analytics worker all call it (Section 18.1). Every result carries a model fingerprint: a hash of the model files and of the calculation code. The fingerprint protects against reuse of an outdated centre calibration; it is not a digital signature of the publisher. A release test suite of 392 tests, including golden-value tests of the September 2026 release, runs on the bundled runtime; a failure of a golden-value test after a change to calculation code is treated as a change of result that requires the owner's decision, not as a broken test.

The platform produces several forms of output: an interactive bilingual (Russian/English) interface; a multi-page PDF report with embedded figures, designed to be archivable as part of the patient's clinical record (the optional narrative of the local LLM is printed separately and is not part of the main PDF); a History analytics CSV built on demand from saved records; and JSON results with 185 output columns for the REST API and the MCP connector. Saved records are encrypted per Windows user, and uploaded clinical tables are never written to disk as separate files.

16.3 Glass-box principle

Every prediction made by the platform can be traced back to a specific coefficient, a specific Monte Carlo iteration, and a specific source of evidence. The GAT layer extends this transparency in a particularly direct way: the clinician can inspect the ten most similar training cohort patients and their documented outcomes, providing a human-interpretable audit trail for the neighbourhood-based prediction. This glass-box property stands in contrast to commercial black-box calculators whose underlying coefficients are inaccessible. By adhering to the principle that high-stakes medical AI must be interpretable and auditable [8,9], every prediction can be explained, critiqued, and improved upon by the wider scientific community.

17. Clinical and Quality-Management Significance

17.1 The platform as a continuous laboratory-surveillance instrument

The principal value of a stage-resolved stochastic prediction model in routine practice extends well beyond individual patient counselling. Because the platform generates a full predicted distribution at every discrete stage of the cycle, it functions simultaneously as a quality-surveillance instrument for the embryology laboratory. Each predicted stage distribution constitutes a quantitative, patient-adjusted expectation against which the corresponding observed value from the culture protocol can be compared in near-real time.

The comparison is not against a fixed population mean but against an expectation conditioned on that specific patient's age, ovarian reserve, and upstream realized values, which removes the principal confounder that renders crude laboratory KPI tracking unreliable: case-mix drift. A laboratory whose patient population shifts toward older or lower-reserve patients will exhibit a falling raw blastulation rate even with unchanged technical performance; the model absorbs this case-mix variation into the prediction itself, so that a deviation between predicted and observed values reflects a genuine change in laboratory or clinical performance rather than a change in the patients presenting for treatment.

17.2 Stage-resolved discrepancy attribution

When aggregate clinical pregnancy rate declines, a single-endpoint predictor can only confirm that performance has fallen. The present platform, by contrast, permits stage-resolved discrepancy attribution. Each stage transition can be audited independently by comparing the observed conditional yield with the model's predicted conditional distribution given the realized upstream value:

If oocyte and maturity counts fall within predicted intervals but fertilization drops below its conditional expectation — the compromised step is localized to the fertilization procedure (sperm preparation, ICSI technique, culture-medium handling). If fertilization and cleavage are concordant with prediction but blastulation falls below expectation — the signal points to the extended-culture environment: incubator gas regulation, medium lot, or oxygen tension. If all embryological stages are concordant but per-transfer pregnancy falls alone — the deviation localizes distally to embryo transfer technique, endometrial preparation, or luteal support. This capacity to identify which link in the chain is responsible converts the platform from a passive performance monitor into an active root-cause-analysis tool, materially shortening the investigation cycle when a non-conformity is detected within a quality-management system.

17.3 The KPIScore as a laboratory variability indicator

The KPIScore is sampled per Monte Carlo iteration and moment-matched to a Beta distribution, so it carries an explicit dispersion as well as a central value. This dispersion is itself diagnostically informative. A widening of the realized KPIScore distribution beyond its predicted dispersion — even in the absence of a shift in its mean — is an early indicator of rising process variability that typically precedes a detectable fall in the mean pregnancy rate. The KPIScore therefore supplies a leading variability indicator in addition to the lagging outcome indicator, allowing intervention before clinical performance is measurably degraded.

17.4 Temporal control charting and dynamic threshold management

Predicted and observed values can be tracked over consecutive time intervals, yielding a continuous, multi-level statistical-control framework. The predicted–observed difference at each stage can be charted in the manner of a Shewhart or cumulative-sum control chart, with the model providing the patient-adjusted center line and Monte Carlo simulation providing the natural control limits. A persistent drift of the observed series below its predicted band constitutes an objective, statistically grounded out-of-control signal, in contrast to arbitrary fixed thresholds that conventional laboratory KPI schemes apply uniformly to all patients.

The control limits are dynamic and stratum-specific: they can be examined separately for clinically distinct subgroups — by age band, by ovarian-reserve category, or by the

unsupervised phenotype cluster to which the patient is assigned — so that a quality signal confined to, for example, the poor-responder stratum is not masked by stable performance in the standard-responder majority. Section 20.3 describes how this framework is implemented in the Early Warning System.

17.5 Tactics for an observed–predicted pregnancy-rate divergence

When the actual clinical pregnancy rate falls below the platform's prediction over a monitored interval, the architecture prescribes a structured escalation rather than an undirected investigation:

1. Vertical decomposition: compare the three-level pregnancy estimate (per-transfer, cumulative-if-viable, overall) against observation to establish whether the deficit arises in the probability of producing a viable transfer or in implantation of transferred embryos.
2. Stage-resolved attribution: isolate the compromised embryological or clinical step as described in §17.2.
3. Stratified analysis: recompute the divergence within phenotype clusters and reserve bands to determine whether the problem is global or confined to a definable subgroup, which materially narrows the differential.
4. Variability inspection: examine the KPIScore dispersion to distinguish a shift in central performance from an increase in process variance, since the two imply different corrective actions.
5. Targeted corrective action and verification: implement and then verify by confirming that the observed series returns within its predicted control band over the subsequent monitored interval, closing the QMS loop with an objective, quantitative endpoint.

17.6 From population extrapolation to validated personalized care

The clinical significance of the platform ultimately rests on a conceptual departure from the dominant paradigm of fertility prognostication. Conventional online calculators return a population-extrapolated point estimate: a probability read from a regression surface fitted to a reference cohort, applied uniformly to every patient sharing a small set of input values. The architecture presented here instead propagates a full probability distribution conditioned jointly on the individual's age, ovarian reserve, realized stage-by-stage values, accumulated attempt history, and the host clinic's own outcome data, and updates this distribution by Bayesian conditioning as each new observation in the cycle becomes available.

The GAT layer introduced in v7.0 adds a further dimension of personalisation: the prediction is not only conditioned on the patient's own parameters but also contextualised by the observed outcomes of real patients with the most similar clinical profiles. The output is therefore not a population analogue but an individualized, dynamically revised forecast specific to the patient and to the treating centre.

18. Integration through the Clinic REST API and the MCP Connector

> ***Key message:*** *In v7.1 every prediction — manual entry, batch table, request from a clinic information system or request from an LLM agent — is produced by the same function with the same models and the same settings. The integration layer transports data and results; it never computes, never applies a separate calibration, and never shows patient identity to an agent. The agent prepares data, launches jobs and explains outputs; the service computes.*

18.1 Single computation path

The desktop service function Service.compute is the only place where a prediction is produced. Its order of operations is fixed: (1) L1–L6 are run, with all models loaded before the random seed is set; (2) the BEFE posterior and the reliability band are computed (L7); (3) the centre intercept calibration is applied (pre-transfer only, Section 7.7); (4) the clinical summary is built, and the cycle probability is re-anchored to the calibrated per-transfer probability. Manual entry in the interface, CSV/XLSX batch calculation, treatment-plan anchoring, calibration exports, the REST API and the analytics worker all call this function. There is no second algorithm and no API-specific calibration: for identical inputs, number of iterations and seed, the analytics API output was verified bit-for-bit against the History analytics CSV of the desktop application.

18.2 Three transports and their trust boundaries

Table 15. *Transports of the v7.1 platform*

Transport	Where it runs	Who uses it	Protection
Desktop server	Loopback only, inside the desktop application	Clinician through the application window	Per-launch session token, strict Host/Origin checks, fixed set of routes, content security policy
Clinic REST API v1	Windows service IVFClinicIntegration71, HTTPS on the clinic LAN (/api/v1)	MIS/LIS and analytics pipelines	TLS ≥ 1.2, loopback or private IPv4 only, exact peer allowlist, API keys stored as SHA-256 hashes, two non-interchangeable key scopes, data encrypted with Windows DPAPI
MCP connector	Local stdio server on the bundled runtime, over the REST API	LLM agents (Claude Code, Codex, Claude Desktop, other MCP clients)	Pseudonymous cycle_uid only; every response passes a privacy filter; crosswalk and keys protected with DPAPI of the current user

18.3 Clinic REST API v1

The REST API is an on-premise interface for a clinic system on the same PC or on a private IPv4 clinic network. It has two key scopes that must not cross. A **clinical** key submits complete, immutable cycle snapshots: the cycle appears in the clinician's "Incoming cycles" inbox, the clinician checks or completes the inputs, and only then is the calculation started;

the result is saved for retrieval by the source system. An **analytics** key submits batches of up to 500 cycles for automatic research scoring: a separate worker process with the models runs at below-normal priority and ends after one idle minute, results never appear in the clinician inbox, centre calibration is not applied, and the clinic prior is pinned per batch and included in the model fingerprint. Results of the analytics scope are labelled scoring = automatic_research and clinician_reviewed = false.

Table 16. *Endpoints of the clinic REST API v1*

Method	Path	Result
GET	/health	ready or license_required; no medical data
POST	/cycles	202 stored; 200 identical duplicate (clinical scope)
GET	/cycles/{id}	Source snapshot and status
POST	/cycles/{id}/requests	Review request; actual state after review
GET	/cycles/{id}/result	202 pending; 200 saved numerical result
POST	/analytics/validate	Same checks as a batch; nothing stored or scored
POST	/analytics/batches	Up to 500 cycles; 202 with counters (analytics scope)
GET	/analytics/batches/{id}	Progress: queued, complete, failed, invalid
GET	/analytics/batches/{id}/results	202 while scoring; 200 with every row
DELETE	/analytics/batches/{id}	Delete batch inputs and results

Data contract. A calculation requires age, AMH, AFC, attempt number, sperm source, and BMI or both height and weight; units are fixed (years, ng/mL, count, kg/m^2, cm, kg). Null or omitted means unknown and zero means observed absence; missing values are never replaced by population means — incomplete snapshots are accepted as needs_data, and in analytics batches the sender imputes explicitly. Numeric strings, non-finite values, unknown fields, inconsistent BMI and increasing known stage counts are rejected. Accepted numeric limits do not imply that the patient is inside the training domain; OOD is assessed during scoring.

Result. prediction.primary equals the desktop headline; kind = per_transfer is a per-transfer probability and kind = current_cycle applies when no transfer is possible. The uncertainty object carries coverage_validated = false; layers describe module availability, not independent calibrated predictions; provenance includes the model and settings fingerprint, application version, calculation time and the hashes of the source and approved snapshots. Saved results are not silently recalculated after a calibration change.

Security and limits. The service opens no outbound connections and uses no cloud or LLM services. At most 25 active keys and 25 allowed peers, 120 requests per minute per IP, 64 KiB per clinical request, five active analytics batches per connection and 50,000 stored analytics rows are enforced; licence expiry blocks medical-data exchange.

18.4 MCP connector

The Model Context Protocol (MCP) [46] connector is a server written with the Python standard library only, running on the runtime bundled with the application and speaking to the local REST API. It is installed as a plugin for Claude Code or Codex (or configured for Claude Desktop and other MCP clients) and supports MCP protocol versions 2024-11-05 to 2025-11-25. With it an agent can inspect clinic tables and MIS exports locally without receiving rows, prepare and validate cycles with the service's own validators, submit jobs

and track progress, send cycles to the clinician's inbox, read all 185 outputs of layers L1–L7 with a built-in interpretation guide, and export results to a local file in which the connector joins the MIS codes back for clinic staff.

Privacy. The agent sees only a pseudonymous cycle_uid (an HMAC of the cycle code with a local secret), the row number of the source file and numeric features. Names, MIS codes, dates, staff names and free text never appear in responses, and every response is checked before it is sent. The cycle_uid ↔ MIS code crosswalk and the API keys are protected with Windows DPAPI; keys are entered in a terminal, never through the chat. Numeric features and forecasts do reach the model provider of the agent, which the clinic must take into account before using the connector with patient data.

Table 17. *Tools of the MCP connector ivf_digital_twin*

Tool	Changes data	Purpose
dt_status	no	Connector version, service and licence state, key validity, recent jobs, setup commands
dt_guide	no	Guide topics (overview, data rules, privacy, interpretation, workflow, errors) and the dictionary of output columns
dt_table_inspect	no	Row count, role and statistics of every column, recognised fields, missing required columns, unit hints — without returning rows
dt_prepare_cycles	local job	Validation with the service rules, rejections by code, imputation flags (only when impute is passed explicitly), service dry run, time estimate
dt_submit	sends	Submits a job; long jobs require confirm_estimated_minutes
dt_job_status	downloads	Progress by state and remaining time (wait ≤ 45 s)
dt_results_overview	no	Descriptive summary: states, timepoints, distributions by layer, reliability, OOD, CSDI, cluster, model fingerprints
dt_results_table	no	Rows by preset (headline, layers, funnel, quality, inputs, risks, retrospective) or by columns, filters and sorting (≤ 100 rows)
dt_cycle_report	no	One cycle by layer group with the meaning of every number and rule-based interpretation notes
dt_export_results	local file	CSV/XLSX for staff with MIS codes joined back locally; content not returned to the agent
dt_delete_job	deletes	Deletes service batches; clinical snapshots remain in the application history

Two timepoints are available. as_entered runs one calculation on the values entered. retrospective runs two: at transfer (all observations, probabilities) and at retrieval (OCC and MII only, funnel forecast), and reports which one was used (transfer, retrieval_fallback or none). Resources expose the guides and the dictionary of the 185 columns; prompts (forecast_table, explain_cycle, what_if) appear as slash commands in Claude Code.

18.5 Rules given to the agent

The connector serves its rules to the agent with every session, and explanatory responses carry answer_requirements that the agent's answer must follow. The rules restate the invariants of this document:

- The service computes; the agent prepares data, runs jobs and explains outputs, after reading the overview and interpretation guides.

- Clinic tables and exports are never opened with the agent's own file tools; paths are passed to dt_table_inspect and dt_prepare_cycles. The only identifier is cycle_uid.
- Blank means unknown and zero means observed absence; nothing is imputed without the user's agreement. MII > OCC means that thawed oocytes were added, not a data error.
- The outcome is clinical pregnancy confirmed by ultrasound (including ectopic), not live birth; clinical_probability is per transfer unless its kind is current_cycle; the range is model uncertainty, not a validated confidence interval; reliability and OOD flags are read with the probability.
- Layers L1–L7 are not independent votes and are never averaged; the final value is BEFE (L7).
- Analytics results are automatic research scoring; clinical cycles are calculated only after clinician review.
- The connector does not compute calibration, KPIs or outcome reconciliation, and the agent does not rebuild them from rows; no diagnoses or prescriptions are issued.

A typical session runs dt_status → dt_table_inspect → dt_prepare_cycles → (user confirms the estimate) dt_submit → dt_job_status → dt_results_overview / dt_results_table / dt_cycle_report → dt_export_results, the last step producing a local file in which clinic staff see forecasts next to their own patient codes.

19. Integration with Laya

> ***Key message:*** *Laya is a fast encoder-classifier, not a generative language model. Fine-tuned on the same training table as the KAN layer, it reaches the discrimination of the DT on external centres and is used beside the DT as a second opinion and as a reader of free-text clinical notes. It is never fused into BEFE: a large disagreement between Laya and the DT is a reason to re-check the cycle, not a vote.*

19.1 System description

Laya (convaiinnovations/laya, Apache 2.0) [47] is an encoder-classifier built on ModernBERT-large (421M parameters) or mmBERT-base (322M) [48]. It receives a state (text or JSON) and typed questions — probability of “yes”, choice or score — and answers in one forward pass (about 33 ms on a GPU and 0.2–0.5 s on a CPU), entirely on the local machine. It was trained with proper scoring rules, but its calibration refers to teacher labels on general tasks, not to clinical pregnancy frequencies. Without fine-tuning it does not read numbers: on 791 transfers its zero-shot output was nearly constant, and when given the DT probabilities it inverted them. All IVF use therefore relies on fine-tuned checkpoints and on a temperature fitted on held-out data.

19.2 Fine-tuning and validation

The tabular checkpoint laya_kan_ft was fine-tuned on the KAN training table with the same split as the KAN layer (12,844 training, 2,569 validation and 643 test rows; 18 KAN features, with features absent in some clinics masked as unknown in 30% of training rows). The text checkpoint laya_text_mt2 (mmBERT) was trained to read clinical notes in Russian and English (Section 19.5).

Table 18. *Laya: development and validation*

Evaluation	Laya	Comparator
KAN hold-out test (643 rows), AUC	0.792	KAN 0.679; gradient boosting 0.820
Four external centres (A–D), weighted within-centre AUC	0.553	DT 0.561; ΔAUC −0.008 [−0.038; +0.024]
Four external centres, pooled AUC after intercept-only centre calibration	0.682	DT 0.682
Centre E 2026, pipeline Laya with centre shift fitted on 2023–2025	0.682	DT 0.702; ΔAUC −0.020 [−0.069; +0.030]
Centre E 2026, Laya fine-tuned on Centre E 2023–2025 (baseline variables)	0.724	DT 0.702; ΔAUC +0.023, p = 0.17

Adding clinical features (AMH, AFC, BMI, sperm source) by a second fine-tuning step on about 640 transfers did not improve discrimination, and at Centre E adding 114 extended clinical variables made Laya slightly worse in fresh transfers (ΔAUC −0.018, p = 0.049). Like every model of the pipeline, Laya reaches a within-centre ceiling of about 0.55–0.62 on these features; it equals the DT but does not exceed it, and its calibration across centres is weaker because it has no clinical inputs from which base rates could be inferred.

19.3 Laya-IVF service: HTTP and MCP

The Laya-IVF service runs the model locally behind an HTTP endpoint on the loopback interface (models laya-ivf for tables and laya-ivf-text for notes) and an MCP server laya_ivf. A centre is selected by an intercept-only logit shift fitted on that centre's outcomes; a new centre needs a few dozen of its own transfers to fit the shift.

Table 19. *Tools of the MCP server laya_ivf*

Tool	Purpose
laya_ivf_status	Model, service state, centre calibrations, limitations, text model
laya_ivf_state	Shows the state that will be sent to the model, input flags, missing required inputs and rule-based restorations, without a prediction
laya_ivf_read_note	Reads up to 100 notes: extracted fields, confidence and the text model's own probability
laya_ivf_assess	Laya probability, centre calibration, flags and comparison with the DT (inputs, text or both)
laya_ivf_note_to_dt	Turns a note into a draft cycle for dt_prepare_cycles, with a status for every field
laya_ivf_assess_dt_job	Scores a finished DT connector job on the same inputs and compares the two estimates

Flags. A value outside the 0.5–99.5 percentiles of the training table is flagged as out of range. If age, attempt number, punctured follicles, retrieved or inseminated oocytes are unknown, or fewer than 6 of 18 features are known, the cycle is marked not assessable. An observed zero of retrieved or inseminated oocytes, 2PN or cleaving embryos makes the per-transfer question inapplicable (the analogue of current_cycle in the DT), while zero blastocysts does not. Missing fields are restored by documented rules only when the user agrees (restore = true), and every restored field is reported. The flag disagreement_high marks cycles in which |logit(Laya) − logit(DT)| after centre calibration of both models falls in the upper third of the distribution observed on Centres A–D (threshold 0.354); in that

third the Brier score of the DT was higher by 0.034 [0.006; 0.064] — a weak but significant signal to re-check the cycle, not a verdict.

19.4 Joint workflow with the Digital Twin

The two connectors work on the same pseudonymous cycles. A free-text note is converted by laya_ivf_note_to_dt into a draft cycle in which every field is marked proposed, needs_review (confidence below 0.9, not included), not_stated, derived by rule, or Laya-only; AMH, AFC, BMI and sperm source are never read from the note and must be asked. The draft goes to dt_prepare_cycles only after a person has confirmed each value. After the DT job is complete, laya_ivf_assess_dt_job reads the same inputs and the DT per-transfer probability from the local job store of the DT connector (no network, no identity crosswalk) and returns both estimates side by side; current_cycle values are not compared. When the number of inseminated oocytes is taken from the DT forecast instead of an observation, the comparison is marked as not independent of the DT.

19.5 Free-text clinical notes

The text checkpoint answers 13 questions about a note — 12 fields (age, attempt, follicles, oocytes, inseminated, 2PN, day-3 cleaving, day-5 embryos, blastocysts, good blastocysts, frozen, transferred) with the option “not stated”, and the pregnancy question. Extracted fields become the inputs of the tabular model, explicit inputs take precedence over the text, and conflicts are flagged. The accuracy of extraction of stated fields was 96.1% (Russian) and 96.3% (English) on hand-written notes and 79.9% and 94.8% on notes written by an independent local LLM; Russian compound numerals written in words remain a known weak point, and errors are often confident, so extracted values must always be shown to the user. Notes must not contain names, MIS codes or dates.

19.6 Laya applications built on the Digital Twin

The same checkpoints are distributed as standalone local applications: Laya Direct 2.0 (one text-and-table model on the loopback interface, with batch tables and nine MCP tools), the Laya Clinic installer for Windows (2.2.0 adds local Whisper dictation; audio is processed in memory and not stored), and an offline Android application “Digital Twin × Laya” (1.3.0) that contains the DT formulas for L1–L3 and the laya_text_mt2 checkpoint in ONNX form and declares no internet permission. In the Android application Laya scores up to 64 weighted representative DT scenarios, and the headline is the weighted median of these probabilities. These applications are research tools; the centre shifts they offer were fitted for laya_kan_ft and their transfer to laya_text_mt2 has not been validated.

20. The Digital Twin in the Laboratory Analytics Pipeline and the Early Warning System

> ***Key message:*** *In the laboratory the DT is not only a counselling tool: it supplies the case-mix-adjusted expectation against which the laboratory's own results are compared. The analytics pipeline uses it as one of several forecasting components; the Early Warning System turns its frozen stage forecasts into risk-adjusted control charts, alongside — not instead of — the absolute thresholds of the Vienna consensus.*

20.1 The analytics pipeline

The laboratory analytics pipeline (embryo-lab-data) processes MIS exports of each clinic through a fixed sequence:

1. Columns are resolved only through a mapping file; outcomes, AMH and BMI are normalised; rows with a transfer and a known outcome are selected for outcome analyses.
2. A local keyed-hash cycle_uid is added; identifiers and names stay in the local layer and never reach agents.
3. Laboratory KPIs of the Vienna consensus [44] are computed as Σ numerator / Σ denominator with 95% CIs, separately for embryologists (FR, ClR, BlR, TGBDR, DegR, 3pNR) and physicians (OCCR, MIIR), together with a separate model KPIScore that reproduces the training formula.
4. Fresh batch results are obtained from the DT 7.1 analytics API (Section 18.3) and from TabPFN, the DNN and Laya; TabPFN-3 regression forecasts the quantitative funnel stages with P10–P90 intervals as an independent QA layer.
5. Calibration, funnel, flags, a review list (priority ≤ 2 or wide interval) and aggregated KPI tables are built locally.
6. For the review list, incubator logs (MIRI) in the culture window, consumable lot changes and, where available, PGT-A reports are examined as candidate explanations, never as automatic causes.
7. Outputs are written atomically with a run manifest, and the status of every component (DT API, TabPFN, DNN, Laya, MIRI, lots, PGT-A) is reported as ok, soft failure with reason, or disabled; zero coverage of an enabled component is treated as a connection failure until proven otherwise.

20.2 Role of DT outputs in the pipeline

Table 20. *How the analytics pipeline uses DT outputs*

DT output	Use in the pipeline	Not used for
Stage medians and intervals (med_, *p025_*, p975_*, CSDI intervals)	Funnel comparison of observed vs expected counts; forecast envelope of the EWS	Judging an individual embryologist without the case-mix context
clinical_probability and BEFE posterior with reliability and OOD	Outcome calibration and review list; comparison with TabPFN, DNN and Laya	Averaging with other models or with other DT layers
Model fingerprint, scoring = automatic_research	Provenance of every run; separation of retrospective and prospective cohorts	Clinical decisions (analytics results are not clinician-reviewed)

Two conventions of the pipeline protect the interpretation of DT outputs. The good-blastocyst stage exists in two bases: the Vienna TGBDR = good blastocysts / 2PN and the model fraction good / blastocysts used by CSDI and by the funnel step Bl→Good; they are bridged by TGBDR = (Bl→Good) × BlR and are never compared directly. Rates above 100% (ClR > 100% after a late fertilisation check; MII > OCC after thawed oocytes are added) are observations, not data errors, and binomial intervals are not computed for them.

20.3 Early Warning System for embryo culture conditions

The EWS follows the design of Bormann et al. [45] — do not wait for the outcome, and monitor the embryologist and the culture conditions in separate loops — but replaces the time-lapse image score of that study with a forecast of the funnel made before culture starts: TabPFN-3 P10–P90 intervals, CSDI intervals and DT medians. The core of the system is risk-adjusted statistical process control on the residuals "observed − forecast": a fall in blastulation explained by a poorer oocyte pool does not raise an alarm, because the forecast already expected it. A second, absolute loop compares the laboratory with the Vienna thresholds, which are imported from the analytics pipeline rather than duplicated.

$$z = (\text{observed} - P50) / ((P90 - P10) / 2.5631), \quad 2.5631 = 2 \cdot \Phi^{-1}(0.9)$$

$$\text{in80} = 1[P10 \le \text{observed} \le P90]$$

The forecast envelope is registered on the day of retrieval (T0) with the model version and run id and is never recomputed: an envelope re-estimated after the result is known is not a forecast. Day 0 and day 1 are registration only; alerts exist only at T+6, the single point of evaluation, and in the draft environment the system raises watch, never alert. Proportion charts use the embryo or oocyte as the unit (hundreds of observations per month instead of tens of cycles), envelope charts use the cycle. In the baseline run on 5,205 observations of four clinics, intervals with P90 − P10 ≤ 1 (12.1% of observations) covered the observed count in only 41.4% of cases versus 81.2% for the others; such degenerate envelopes are excluded from the residual layer, and the measured coverage by stage — not the nominal 80% — is the reference. A fall in coverage at every stage at once, including stages before culture, is not treated as a culture signal: it is examined in a fixed order as a change of case mix, model drift or a laboratory-wide factor. Incubator logs and lot changes can raise watch to alert but never create or retract an alert.

20.4 Division of labour

Table 21. *Division of labour between the DT, the analytics pipeline and the EWS*

Question	Answered by
What is this patient's probability of clinical pregnancy, and how certain is it?	DT 7.1 (BEFE), reviewed by the clinician; Laya as a second opinion
What did the laboratory produce at each stage, given this patient's case mix?	DT stage forecasts and TabPFN-3 intervals, frozen at T0
Is the laboratory below an absolute standard?	Vienna KPI tables of the analytics pipeline
Has the laboratory fallen relative to itself, early enough to act?	EWS residual charts at T+6
Why did a stage deviate?	Stage-resolved attribution (§17.2) with MIRI logs, lots and PGT-A as candidate confounders

20.5 Limitations

The discriminative metrics of the models (Brier score, AUC) are not part of the EWS interpretation: at the scale of a single laboratory they cannot be distinguished from noise. The stage biases of the DT at a given centre (Section 14.8.4) mean that the envelope must be established on that centre's own history before residuals are interpreted. The EWS is in development: the infrastructure and baseline measurements are complete, and the historical backfit and calibration of alarm thresholds precede its use in the laboratory.

21. Conclusions

The IVF Digital Twin v7.1 is, to our knowledge, the first IVF prediction system to integrate stochastic stage-wise simulation, multi-source per-transfer ensemble, calibrated neural-network prediction, Bayesian evidence synthesis with clinic-specific data, mid-cycle conditional updating, unsupervised phenotype classification, diffusion-based generative modelling, patient-similarity graph analysis, and formal multi-source Bayesian fusion with reliability scoring within a single transparent, auditable framework.

The system addresses several gaps that conventional point-estimate calculators leave open: it propagates full probability distributions throughout the prediction pipeline; it decomposes pregnancy probability into per-transfer, cumulative-if-viable, and overall cycle success levels; it quantifies OHSS and empty-cycle risk explicitly; it allows mid-cycle observations to re-condition all predictions; it provides an independent data-driven generative verification (L5) from approximately 15,000 real cycles; it contextualises the prediction within the outcomes of clinically similar real patients (L6); and it formally integrates all upstream signals into a single posterior probability with an uncertainty range, OOD detection, and reliability scoring (L7).

The BEFE layer (L7) represents the architectural completion of the IVF Digital Twin. Where earlier versions produced several probability estimates from different methodological perspectives without a principled means of combining them, v7.0 formally integrated these into a single uncertainty-quantified output whose provenance is fully auditable through the fusion-pull decomposition. When all layers agree strongly, the clinician receives a tight range and high reliability score. When layers disagree, the range widens proportionally, and the OOD detector provides diagnostic guidance about which dimension of the patient's profile is unusual. The meta-question “how confident should we be in this answer, and why?” is a first-class output alongside the prediction itself.

Version 7.1 consolidates this architecture rather than extending it. The mechanistic layer now rests on count distributions refitted on local clinical data and is conditioned exactly on every observation; transfers of one cycle are correlated; and the uncertainty range narrows as the cycle is observed while still widening when the sources genuinely disagree. The same computation is available to clinicians, to clinic information systems and to LLM agents through one code path, with patient identity kept on the clinic's machine. In the prospective in silico validation at Centre E the frozen model discriminated as well as any model of the pipeline that had not seen the centre's data, and its calibration was corrected by a centre recalibration fitted on earlier years only — the behaviour expected of a model that is designed to be adapted locally rather than refitted.

Together, the seven layers provide a multi-perspective, multi-methodology assessment: Monte Carlo stochastic simulation (L1), ensemble pregnancy scoring (L2), neural network prediction with Bayesian clinic calibration (L3), cluster-based profiling (L4), diffusion-based laboratory forecasting (L5), patient similarity graph analysis (L6), and formal evidence fusion (L7). Each layer contributes an independent signal; each is fully auditable and reproducible; and their formal integration in L7 ensures that the final output is not merely the loudest individual voice but a principled synthesis of all available evidence.

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